

Improve the Charge Carrier Transporting in Two-Dimensional Ruddlesden–Popper Perovskite Solar Cells

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Conventional 3D organic–inorganic halide perovskite materials have shown substantial potential in the field of optoelectronics, enabling the power conversation efficiency of solar cells beyond 26%. A key challenge limiting the further commercial application of 3D perovskite solar cells is their inherent instability over outer oxygen, humidity, light, and heat. By contrast, 2D Ruddlesden–Popper (2DRP) perovskites with bulky organic cations can effectively stabilize the inorganic slabs, yielding excellent environmental stability. However, the efficiencies of 2DRP perovskite solar cells are much lower than those of the 3D counterparts due to poor charge carrier transporting property of insulating bulky organic cations. Their inner structural, dielectric, optical, and excitonic properties remain to be primarily studied. In this review, the main reasons for the low efficiency of 2DRP perovskite solar cells are first analyzed. Next, a detailed description of various strategies for improving the charge carrier transporting of 2DRP perovskites is provided, such as bandgap regulation, perovskite crystal phase orientation and distribution, energy level matching, interfacial modification, etc. Finally, a summary is given, and the possible future research directions and methods to achieve high-efficiency and stable 2DRP perovskite solar cells are rationalized.

rapid development.^[4,5] The power conversion efficiency (PCE) of the perovskite solar cells (PSCs) has increased from 3.8% to 26.1% in just a few years.^[6] However, 3D PSCs still suffer from severe inherent instabilities over oxygen, humidity, light, and heat, limiting their further commercialization.^[7–9] To enhance the stability, reducing the dimension from the 3D perovskite phase to the low-dimensional Ruddlesden–Popper phase has been proven as an effective way by incorporating bulkier organic ammonium molecules into the perovskite lattice.^[10,11]

The bulky organic cations in two-dimensional (2D) perovskites are enabled to stabilize the inorganic slabs and prevent the inorganic perovskite layers from oxygen and moisture invasion. The inorganic framework holds great promise for optoelectronic properties, whereas organic cations play a role in many molecular structures and other physical possibilities.

According to the crystal orientation of 3D perovskites, it can be cut along three crystallographic directions $\langle 100 \rangle$, $\langle 110 \rangle$, and $\langle 111 \rangle$ to obtain different multilayer structures,^[12] in which $\langle 100 \rangle$ orientation layered perovskite is the most common orientation. In the transformation from 3D perovskites to 2D perovskites, partial A-site cations in 3D perovskites are replaced by bulky organic cations.^[13] The 2D perovskite structures are mainly classified into two categories. One is the Ruddlesden–Popper (RP) phase perovskite with a general formula of $A'_2A_{n-1}M_nX_{3n+1}$, where A' is generally monovalent aliphatic or aromatic alkylammonium cations (e.g., *n*-butylammonium (*n*-BA⁺), 2-phenylethylammonium (PEA⁺)). The other is the Dion–Jacobson (DJ) phase perovskite with a general formula of $A''A_{n-1}M_nX_{3n+1}$, where A'' is generally divalent cations. In the formula, A represents an organic cation, such as a methylamine cation (CH_3NH_3^+ , MA⁺) or formamidinium cation ($\text{CH}_2(\text{NH}_2)_2^+$, FA⁺), M is a divalent metal cation, X is a halide amino, and n is the number of $[\text{MX}_6]^{4-}$ octahedral layers between two organic insulating layers.^[14] The most commonly employed 2D perovskite structure is RP phase perovskite due to its much higher structural versatility. In the 2DRP perovskite crystals, the Van der Waals force of the bulky spacer bilayer structure and the hydrogen bonds between organic spacers and inorganic layers maintain the structural integrity.^[15] Earlier studies have shown that 2DRP perovskites exhibit apparent advantages due to their unique

1. Introduction

Organic–inorganic halide perovskite ($\text{CH}_3\text{NH}_3\text{PbX}_3$, $X = \text{Cl}, \text{Br}, \text{I}$) materials have attracted much attention recently.^[1–3] Photovoltaic devices using three-dimensional (3D) perovskites as the active layer material have experienced unprecedented

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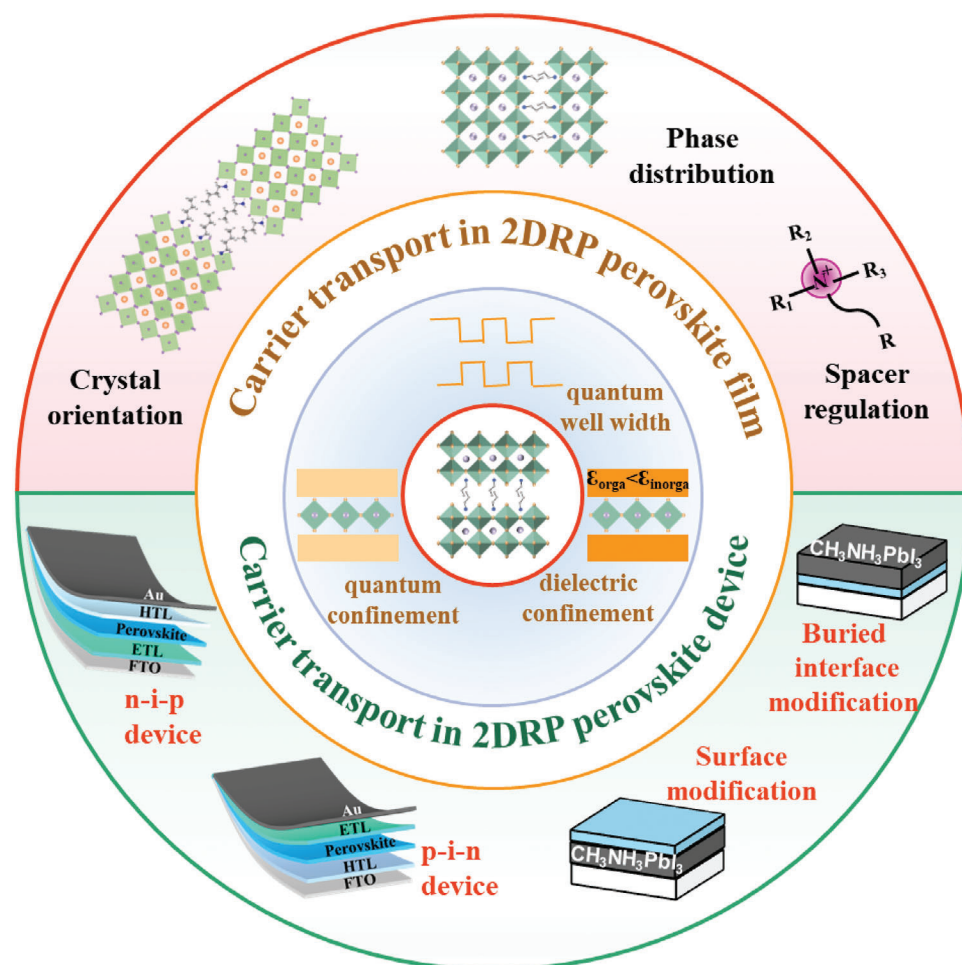


Figure 1. The origin and solution of carrier transporting property in 2DRP PSCs.

properties, such as high formation energy, suppressed ion migration, reduced carrier recombination, low self-doping effect, etc.^[16,17]

2DRP perovskite structures can be considered as multiple quantum wells, and the charge carriers are trapped within the inorganic layer and cannot pass through the insulating bulky organic cations due to the quantum confinement of the structure. Therefore, the efficiency of 2DRP solar cells usually suffers from the poor charge transporting characteristic, which is much lower than that of 3D counterparts. Achieving efficient charge-carrier transporting in 2DRP PSCs is a critical issue. Despite the vast progress of 2DRP PSCs, very few reviews on improving the carrier transport of 2DRP have been reported. A reasonable perovskite bandgap plays a crucial role in the performance of PSCs. Perovskites are used as the light-absorbing layer in solar cell devices, and the narrow bandgap contributes to absorbing more incident light and yielding more significant numbers of charge carriers. Herein, we provide a comprehensive summary for achieving efficient charge transport in 2DRP perovskite thin films and devices, as shown in **Figure 1**. Firstly, approaches to adjusting the perovskite bandgap have been discussed, including n values, A cation and halogen anion regulation. Moreover, how to manipulate perovskite crystal lattice and improve the charge carrier

transport in 2DRP perovskite films are analyzed in detail, such as the regulation of perovskite crystal phase orientation and distribution and control of organic spacer cation. Apart from the carrier transport in the perovskite films, keeping both electrons and holes in the external circuit along their respective transmission paths is crucial for the overall performance of PSCs. Here, selecting the appropriate carrier transport layers to match the energy level and modifying the functional interfaces to help carriers pass through are discussed to promote the integral carrier transport inside the PSCs. Finally, a summary is given, and the possible future research directions and methods to achieve high-efficiency and stable 2DRP perovskite solar cells are rationalized.

2. The Fundamental Properties of 2DRP Perovskites

The transformation along the $\langle 100 \rangle$ orientation from 3D perovskites to 2D perovskites is shown in **Figure 2a**. Because of the introduction of organic cations, the 2DRP perovskite structure can be considered multiple quantum wells, where the inorganic layers act as wells, and the organic spacer layers act as barriers. Due to the quantum confinement of the structure, excellent physical and chemical properties of 2DRP

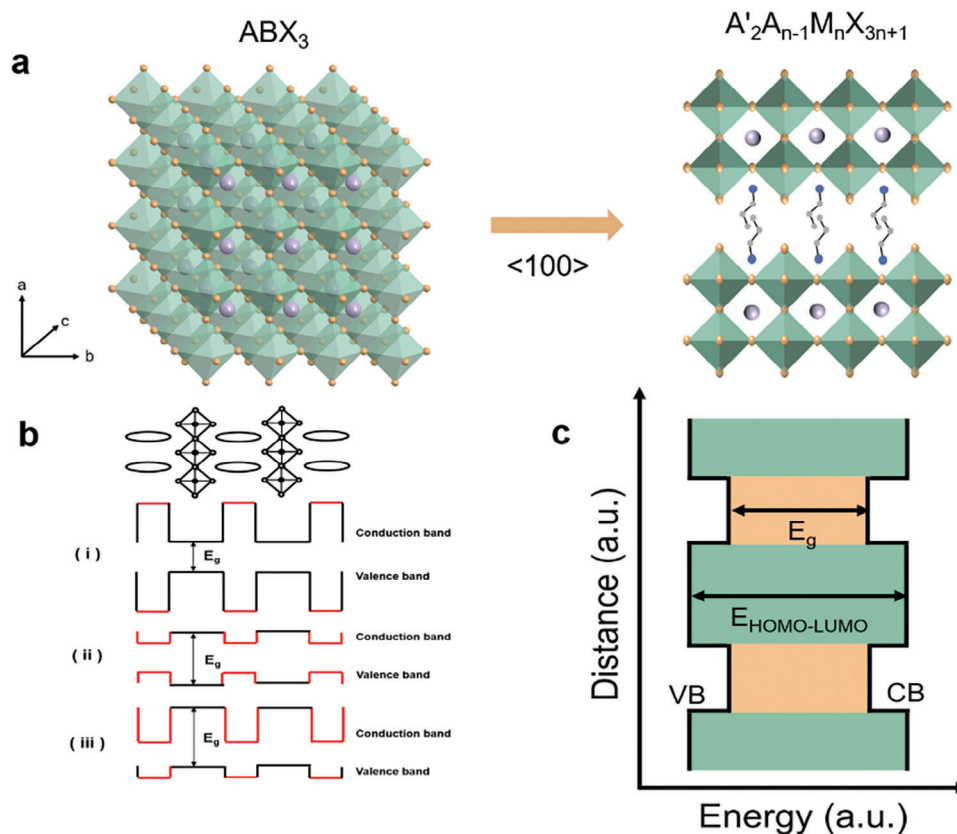


Figure 2. a) 2D layered perovskites derived from the basic 3D AMX_3 structure by taking cuts from $\langle 100 \rangle$ direction. b) Schematic organic–inorganic hybrid perovskite structures and several possible energy-level schemes can arise within these structures. Reproduced with permission.^[19] Copyright 2018, American Chemical Society. c) Energy diagram corresponding to the 2D structure in panel b(i). Reproduced with permission.^[21] Copyright 2018, American Chemical Society.

perovskite, including the suppressed ion migration, high formation energy, low self-doping effect, and reduced carrier recombination, can be realized.^[16,17] The band structure of 2D perovskite is shown in Figure 2b–i.^[18] The organic layer shows a wider bandgap than the inorganic layer in the layered hybrid organic–inorganic perovskite, and it was originally proposed by Mitzi as a type-I quantum well^[19] and is verified in the $(C_{10}H_{21}NH_3)_2PbI_4$ perovskite.^[20] As shown in Figure 2c, this configuration's inorganic layer (yellow regions) acts as a potential well, and the organic layer (green regions) as a potential barrier.^[21] In the case of the conduction band (CB), when electrons move from higher energy to lower energy, the energy is released. If the energy difference between the two positions is large, it is difficult for the electrons to absorb enough energy at once to cross the barrier, trapping them in the well.

Suppose the width of the potential well is reduced to the carrier's de Broglie wavelength. In that case, the charge's behavior will be constrained at the quantum level, forming a typical quantum well structure.^[22] Of course, the structure of the quantum well will change slightly depending on the material. Figure 2b–ii belongs to the type-I quantum well structure, but it differs from Figure 2b–i because the inorganic and organic layers play opposite roles. In this structure, the CB of the inorganic layer is higher than the organic layer's lowest unoccupied molecular orbital (LUMO). In comparison, the valence band is lower than the

highest occupied molecular orbital (HOMO) of the organic layer, resulting in the inorganic layer becoming the barrier and the organic layer becoming the potential well. The structural changes shown in Figure 2b–iii are more obvious than the first two structures. While the CB of the inorganic layer is higher than the LUMO orbital of the organic layer, the valence band is also higher than the HOMO orbital, and a type-II quantum well structure is formed. Inevitably, the existence of a quantum well will greatly limit the transmission of charge carriers in perovskite and then reduce the current.

2.1. Quantum Confinement Effect

When the n value is small, the thickness of 2DRP phase perovskite is close to the de Broglie wavelength of the carrier, resulting into quantum confinement effect on the charge carriers. The quantum confinement effect is one type of fundamental phenomenon in the field of semiconductors. Just as the surface states of matter are different from the interior, the material's physical properties are considerably changed when its size is equivalent to the Bohr radius of the exciton.^[23] In 2019, Parrott et al.^[24] studied the quantum confinement of ultra-thin $MAPbI_3$ films by the vacuum evaporation technology. They found that the thicknesses of the inorganic layer corresponding to n values from 1 to 4 are

0.6, 1.3, 1.9, and 2.6 nm, respectively, much lower than the Bohr radius (≈ 5 nm). The thickness decrease of inorganic layer enhances the quantum confinement effect significantly. Typically, the quantum confinement effect is particularly pronounced in type-I quantum wells, whose electrons and holes are confined to the same material, resulting in stronger binding on exciton. Hong et al.^[22] showed that the Bohr radius was halved, and the exciton binding energy was increased due to the quantum confinement effect in PbI_4 -based layered semiconductors. Therefore, increased exciton binding energy and bandgap usually accompany the quantum confinement effect.

2.2. Dielectric Confinement Effect

Dielectric confinement is more common than the quantum confinement effect, and the structure of 2DRP phase perovskite determines it. Organic macromolecules and inorganic octahedrons usually appear alternately in 2D crystals, while the dielectric constant of the organic layer is low and that of the inorganic layer is high. The massive difference of these two structures will increase the exciton binding, leading to the stable existence of excitons at room temperature and their confinement in the internal electric field composed of the positively charged organic layer and the negatively charged inorganic layer, unfavorable to free carrier charge dissociation efficiency.^[25] Traore et al.^[18] applied the image-charge and first-principle methods to hybrid organic-inorganic perovskites to explain the dielectric confinement effect. As shown in **Figure 3a**, they calculated the high-frequency dielectric constants along the stacking axis at the composite well/barrier interface, perovskite well, and organic barriers. They compared the composite interface's dielectric profile with the sound barrier's combined dielectric curve, and elucidated the effects of dielectric confinement on the energy band of inorganic wells, the barrier thickness of quantum wells, and so on.

2.3. Charge Carrier Transporting in Quantum Wells

It is found that the vast majority of carriers in 2DRP perovskites exist in the form of excitons,^[30] and there are various views on how the excitons dissociate into free carriers and the working mechanism of the carriers in perovskites. Blancon et al.^[26] showed that the excitons undergo the decay of the classical process after their generation and thus have a very short lifetime in perovskites with $n \leq 2$. However due to the presence of "layer-edge states" at the interface in perovskites with $n > 2$, the excitons can dissociate into free carriers trapped in the layer-edge states (**Figure 3b**). Such carriers are not affected by non-radiative decay mechanisms such as electron-phonon coupling, and thus have long lifetimes. On this basis, Zhao et al.^[28] proposed a new mechanism of trap-enabled long-distance carrier transport. When an exciton partially dissociates into free electrons and holes, the holes are rapidly trapped, while the electrons migrate rapidly to the low-energy sites in the conduction band, and this carrier transport characteristic with different migration speeds will be conducive to longer transport distances (**Figure 3d**). After that, Yin et al.^[29] proposed an Auger-assisted electron transport mechanism. When the exciton band edge energy is similar

to the electron energy barrier, the electrons can obtain enough energy to cross the potential barrier of quantum well through the Auger recombination, thus realizing the charge carrier transport among perovskite layers (**Figure 3e**). They later experimentally confirmed this mechanism in 2D single crystals. In addition, the self-charge-separation property of the 2D perovskites films with rational quantum-wells energy level arrangement is conducive to the extraction and charge carrier transport. Liu et al. studied BA-based 2D perovskites thin films by transient-resolved PL spectroscopy and absorption spectroscopy indicate that the films are ordered from bottom to top as an ordered array of phases with minor to large n values, and that such an energy-band arrangement makes the electrons and holes are transferred in opposite directions. Each accumulates on the upper and lower surfaces of the perovskites (**Figure 3c**).^[27] Furthermore, the electron-phonon coupling phenomenon and self-trapped excitons generated by lattice elastic distortion are widely shown in 2D perovskite quantum well structures. The phonon scattering of charge limits the carrier mobility to a certain extent while leading to the formation of large polaritons,^[31] and such physical processes of phonon participation in exciton dissociation and charge transport also need to be further investigated.

2.4. Constrained Charge Carrier Transport

From the above analysis, it can be seen that the factors affecting charge transport in the current 2DRP perovskite devices are manifold. In general, it mainly contains the following three points.

Firstly, 2DRP perovskite crystals tend to be anisotropic due to the presence of quantum well structures. When different organic amine molecules are introduced, the inorganic octahedra within the RP phase undergo different degrees of distortion in the three crystallographic directions,^[32] and the expansion and contraction of the crystal lattice and the relative displacement of the atoms further enhance the asymmetry of the crystals. In fact, the anisotropic crystal structure is directly related to the random orientation of the phases. Different crystal planes in the perovskite crystal tend to grow with different atomic arrangements, resulting in a large number of randomly oriented domains inside the crystal,^[32] which become obstacles on the charge transport channel.

Secondly, due to the self-assembly effect of perovskite crystals, a chaotic distribution of phases with different n values, i.e., the coexistence of multiple quantum wells, is presented in most 2DRP perovskite films. This implies that the degree of quantum confinement inside the crystal is unevenly distributed, which, together with the presence of organic insulating molecules, makes the charge carrier transport between different quantum wells even more inefficient.

Thirdly, due to the dielectric confinement effect, 2D perovskites produce electrons and holes with strong Coulomb interactions after absorbing photons, and they are attracted to each other to form excitons. Due to the quantum confinement effect, the excitons can be stabilized at room temperature. In contrast, the unique structure of the quantum well makes it difficult to cross the potential barrier of the organic layer. Thus, the excitons can only be confined in the inorganic layer, which makes the 2DRP perovskites have exciton binding energies as high as a

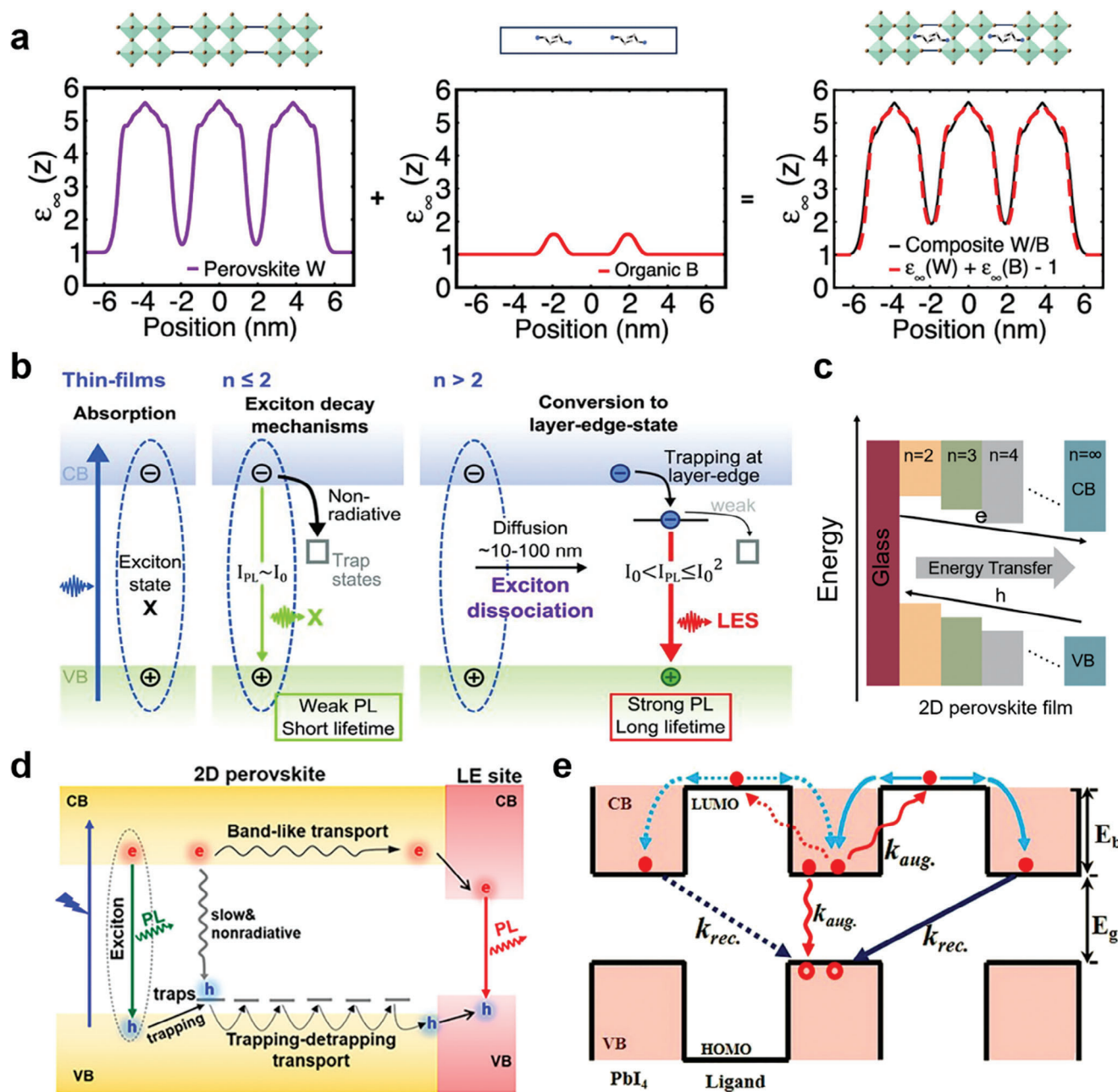


Figure 3. a) Composite approach to dielectric profiles and band alignment. Reproduced with permission.^[18] Copyright 2018, American Chemical Society. b) Summary of the main photoemission mechanisms in thin films. Reproduced with permission.^[26] Copyright 2018, AAAS. c) Comparative bandgap energy alignment of $(BA)_2(MA)_{n-1}Pb_nI_{3n+1}$ perovskites with different n values. Three possible carrier transfer mechanisms, electron and energy transfers from small- n to large- n perovskites and hole transfer from large- n to small- n perovskites, are all energetically allowed. Reproduced with permission.^[27] Copyright 2017, American Chemical Society. d) Schematic of a trap-mediated transport mechanism in 2D layered perovskites. Reproduced with permission.^[28] Copyright 2020, American Chemical Society. e) Proposed Auger-assisted electron transfer and recombination between neighboring perovskite QWs with $E_g \approx E_b$ energy alignment. The Auger recombination ($k_{aug.}$) of an exciton can pump a second electron to the LUMO level of the ligand, and the electron can transfer back either to the original QW or to the neighboring QW. The separated electron and hole can recombine ($k_{rec.}$) by tunneling through the ligand layer. Reproduced with permission.^[29] Copyright 2021, American Chemical Society.

few hundred meV. The extremely strong exciton binding energy leads to the difficulty of exciton dissociation and the lack of free charge carriers. Thus, the 2DRP perovskite material is extremely limited in charge carrier transport and performs poorly when applied to photovoltaic devices.

3. Bandgap Regulation

2DRP perovskites as a light-absorbing layer have excellent stability but incomplete light absorption. Although the 2DRP phase perovskite materials have larger bandgaps than the

corresponding 3D compositions, their bandgap can be adjusted differently. It is shown that the bandgap of perovskites is mainly determined by the thickness of the quantum well and is also affected by the ionic species of the composition.

3.1. n Value Regulation

In a 2D perovskite structure, the quantum well width is mainly related to the n value. The n value determines the proportion of large organic cations to A-site cations. The smaller the n value, the more significant the proportion of large organic cations. $n = 1$ represents the condition of monolayer 2D perovskites,^[33] and $n = \infty$ represents the crystal appearing as a long-range ordered octahedral cubic structure with the general formula of ABX_3 , and the macroscopic structure becomes three-dimensional. The $n = 1$ layered perovskite without A-site cations has a wide bandgap. The large A' ions lead to the deformation of the inorganic layer and thus affect the crystal bandgap. Tanaka et al.^[34] systematically studied the electronic and exciton structures of $(C_4H_9NH_3)_2PbBr_4$ crystal with $n = 1$. They found that the excitons in $(C_4H_9NH_3)_2PbBr_4$ are Wannier-type. The 2D space constraint and the like charge effect enormously enhance the exchange interaction between electron and hole, leading to the exciton splitting and significantly increasing the exciton binding energy.

By contrast, 2D perovskites with $n > 1$ are commonly used. As early as 2003, Tanaka et al.^[23] studied the optical and electrical absorption of lead-halide-based quantum-well crystal group $(C_6H_{13}NH_3)_2(CH_3NH_3)_{n-1}Pb_nI_{3n+1}$ ($n = 1, 2, 3$). They found that the exciton resonance energy exhibits blue-shifted with the decrease of the potential well thickness (corresponding to n value) and the bandgap increase. In contrast, the binding energy of the exciton increases sharply due to the space constraint and the image charge effect. As $n \geq 2$, the crystal's bandgap can be estimated by effective mass approximation, although it is unsuitable for the crystal with $n = 1$. In 2014, Smith et al.^[35] prepared 2D $(PEA)_2(MA)_2[Pb_3I_{10}]$ perovskite to reduce the moisture sensitivity of 3D lead perovskite. They obtained a device with an open circuit voltage (V_{OC}) of 1.18 V and power conversion efficiency (PCE) of 4.73% for the first time. Since then, 2D perovskites with different n values have attracted wide attention. Stoumpos and co-workers^[36] studied the crystal structure of 2D perovskites $(BA)_2(MA)_{n-1}Pb_nI_{3n+1}$ with n values of 2, 3, and 4, respectively. The 2DRP perovskites with different n values have different solubility. The precipitation of compounds at room temperature showed obvious color difference (Figure 4a). The structural characterization of the crystal precipitates revealed an underlying tendency that compounds with odd-numbered layers ($n = 1, 3$) adopt a higher symmetric configuration than those with even layers ($n = 2, 4$). The samples' room-temperature absorption and PL spectra were further carried out. As shown in Figure 4b, sharp optical absorption edges could be seen in the spectrum, indicating the direct bandgap trend of the crystal. The energy of bandgap (E_g) increased from 1.50 eV ($n = \infty$) to 2.43 eV ($n = 1$) with the decrease of n value. The calculation of density functional theory (DFT) suggests that when the number of perovskite layers increases to $n = 3$ and $n = 4$, the stability of the non-centrosymmetric phase is significantly higher than that of

the centrosymmetric structure, that is, for the perovskite with n value of 4, the non-centrosymmetric structure is extremely favorable to the stability.

The change of bandgap is directly related to the exciton binding energy. In order to explain the intrinsic relationship between bandgap and n value, Blancon et al. conducted a quantitative analysis of the exciton binding energy of $(BA)_2(MA)_{n-1}Pb_nI_{3n+1}$ perovskites. As shown in Figure 4c, with the increase of n value, the optical bandgap in the visible spectrum is tuned from 2.540 ± 0.004 eV (488 nm) ($n = 1$) to 1.846 ± 0.004 eV (672 nm) ($n = 5$), and the absorption is enhanced. As the quantum well thickness increases from $n = 1$ to 5, the exciton mass decreases from $0.221 m_0$ to $0.186 m_0$ (m_0 is the free electron mass). The exciton binding energy decreases from 470 to 125 meV, consistent with the change law of E_g .^[37] It is worth noting that the relationship between stability and bandgap change with n value is opposite. Quan et al. adopted DFT to prove that perovskite with low formation energy can easily degrade with poor stability. As shown in Figure 4d, the formation energy of 2DRP perovskite decreases with the increase of the n value, and the stability decreases accordingly.^[17] Many studies suggest the bandgap and stability relationship of 2DRP perovskite can be more balanced when $n = 4$ or 5, more favorable to the overall performance of perovskite devices.^[40,41]

3.2. A Cation Regulation

Tolerance factor (t) was introduced to describe the structural stability of perovskites, and $t = (R_A + R_X)/\sqrt{2}(R_B + R_X)$, in which R_A , R_B and R_X are the ionic radii of ions at A, B, and X sites, respectively. The tolerance factors of stable perovskite structures are generally between 0.78 and 1.05.^[42] Organic cations play the role of lattice charge compensation in organic-inorganic hybrid perovskite materials, and their size will cause the expansion and contraction of the lattice, affect the bond length between metal ions and halogen ions, and further affect the crystal bandgap.^[43,44] Three major A-site cations in 2DRP phase perovskites can meet the tolerance factor condition, that is, MA^+ , FA^+ , and Cs^+ .

The perovskite $MAPbI_3$ is a direct bandgap semiconductor with a bandgap width of about 1.55 eV, and its high absorption coefficient and low exciton binding energy ($\approx 50 \pm 20$ meV) contribute to preparing high-performance PSCs.^[43,45,46] As a result, initial 2DRP perovskite studies also focused mainly on the MA group. The most common $BA_2MA_2Pb_3I_{10}$ film has an optical bandgap of 1.89 eV,^[47] which is higher than that of 3D films of the same composition, resulting in poorer light absorption and inferior performance to 3D.

As the study progressed, the researchers found that MA^+ -based perovskite materials often have poor thermal and humidity stability due to the volatility of MA^+ . Cui et al.^[48] replaced MA^+ with a larger radius FA^+ to prepare perovskite for photovoltaic devices. The tolerance factor of FA^+ is slightly greater than 1,^[49] the lattice expands, and the bandgap decreases due to the increase of ionic radius.^[50] The bandgap of $FAPbI_3$ is 1.41 eV,^[51] closer to the optimal bandgap of 1.4 eV for the light-absorbing materials. After replacing the MA^+ cations in $(ThMA)_2(MA)_4Pb_5I_{16}$ perovskite films with FA^+ , Lai et al. found that a narrower bandgap

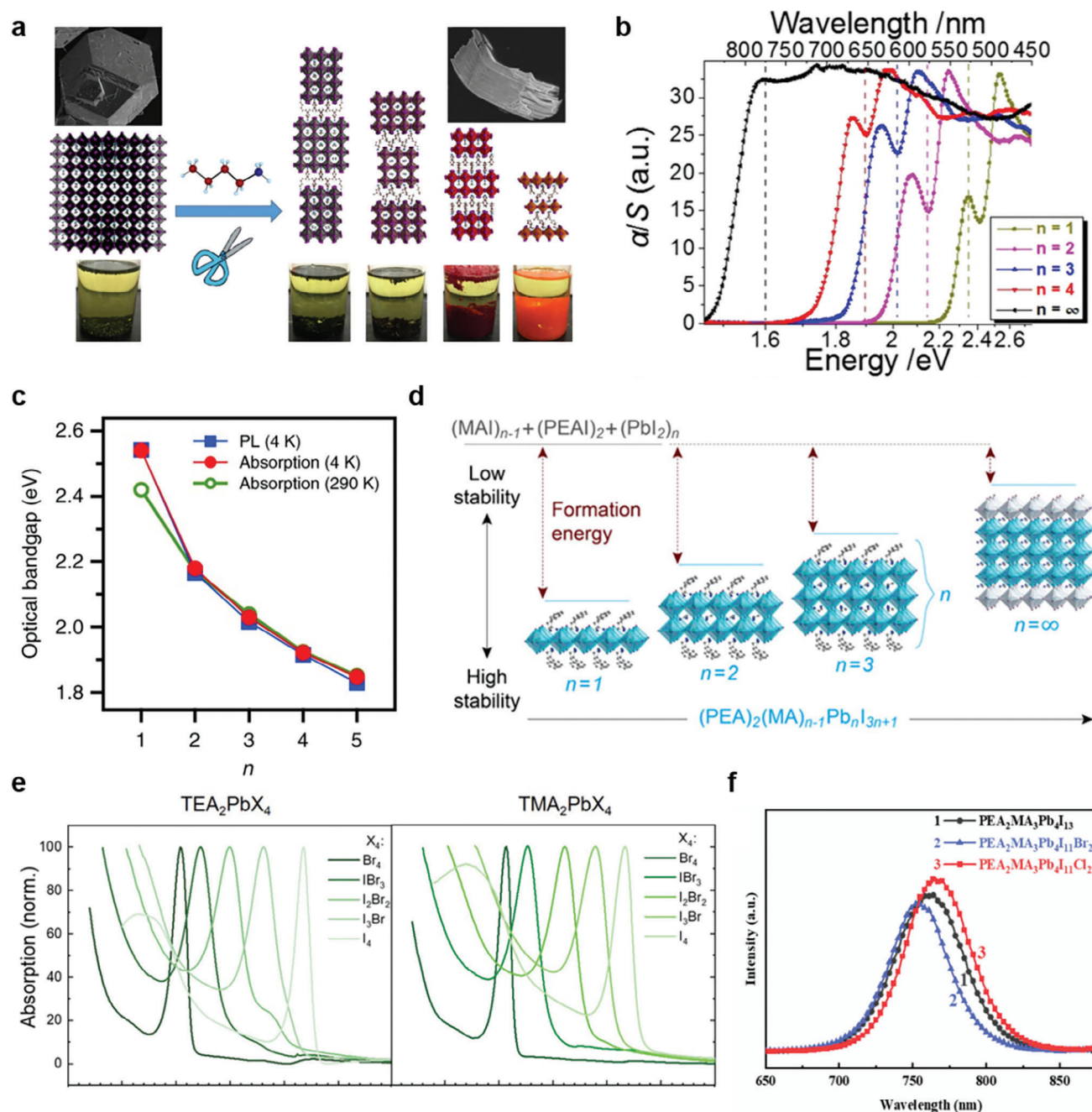


Figure 4. a) SEM images (top) and photographs of the $(\text{BA})_2(\text{MA})_{n-1}\text{Pb}_n\text{I}_{3n+1}$ perovskite crystals (bottom) (scale bars = 200 μm). Reproduced with permission.^[36] Copyright 2016, American Chemical Society. b) Optical absorption of polycrystalline samples obtained from diffuse reflectance measurements converted using the Kubelka–Munk function ($\alpha/S = (1 - R)^2/2R$). Reproduced with permission.^[36] Copyright 2016, American Chemical Society. c) Experimental optical bandgap scaling. Reproduced with permission.^[37] Copyright 2018, Springer Nature. d) Energetics of perovskite formation and stability. The unit cell structure of $(\text{C}_8\text{H}_9\text{NH}_3)_2(\text{CH}_3\text{NH}_3)_{n-1}\text{Pb}_n\text{I}_{3n+1}$ perovskites with different n values, showing the evolution of dimensionality from 2D ($n = 1$) to 3D ($n = \infty$). Reproduced with permission.^[17] Copyright 2016, American Chemical Society. e) The UV–vis absorption spectra of TEA- and TMA-based I–Br perovskite materials. Reproduced with permission.^[38] Copyright 2021, The Authors, published by Wiley-VCH. f) PL spectra of $\text{PEA}_2\text{MA}_3\text{Pb}_4\text{I}_{13}$ film, $\text{PEA}_2\text{MA}_3\text{Pb}_4\text{I}_{11}\text{Br}_2$ film, and $\text{PEA}_2\text{MA}_3\text{Pb}_4\text{I}_{11}\text{Cl}_2$ film. Reproduced with permission.^[39] Copyright 2021, Elsevier.

contributed on extended light absorption spectrum, and the device photovoltaic performance was significantly improved. At the same time, the FA-based perovskite devices exhibit better thermal stability.^[52,53] Similarly, Shao and co-workers prepared a series of $(4F\text{-PEA})_2(\text{FA}_x\text{MA}_{1-x})_4\text{Pb}_5\text{I}_{16}$ ($n = 5$) 2D PSCs with different FA substitution ratios, pure MA ($x = 0$), pure FA ($x = 1$) and FA/MA mixture ($x = 0.3$). They found that pure FA-based perovskite has a unique oblique crystal orientation and a more extensive light absorption range, which is beneficial to improving the device's performance.^[54]

Snaith et al.^[55] introduced a smaller radius Cs^+ into perovskite and prepared all inorganic perovskite CsPbI_3 . The stability of Cs-based perovskite was significantly enhanced. However, its tolerance factor is only about 0.8 due to the small radius of Cs^+ ,^[49] leading to severe lattice shrinkage, short bond length between metal ions and halogen ions, and an increased bandgap in perovskites.^[43] Zhang et al.^[56] doped Cs^+ into $\text{BA}_2\text{MA}_3\text{Pb}_4\text{I}_{13}$ films and found that the lattice spacing was altered due to the partial substitution of Cs^+ for MA^+ , and accordingly, the quality of the films and the carrier dynamics were altered to some extent. The high quality of the films and the improvement of the trap density and carrier mobility resulted in a significant enhancement of the device optoelectronic performance. The doping of inorganic Cs^+ more significantly enhances the tolerance of the material to high humidity and sustained thermal stability.

Overall, most of the current studies have chosen FA^+ as the A-site cation in order to obtain superior photoelectric performance, but FA-based perovskites need to be regulated by certain means due to the possible phase transitions at different temperatures.^[57] The current performance of Cs-based 2DRP perovskite devices has yet to be broken through, but the extremely high stability of Cs^+ makes it expected to be the focus of future research. Of course, more efficient and stable 2DRP PSCs may also be realized if different cations doping or direct selection of mixed cations is carried out to make full use of the advantages of all three.

3.3. Halogen Anion Regulation

Halide anion substitution in 2DRP perovskites can also be used to modulate the bandgap. Similar to the 3D perovskite, changing the halide anion from I^- to Br^- or Cl^- will promote a blueshift in the optical properties. Zanetta et al.^[38] explored a series of 2DRP perovskites with different halide substitution using thiophene-based long chain organic cation 2-thiophene ethyl ammonium (TEA) and 2-thiophene methyl ammonium (TMA). As shown in Figure 4e, by increasing the content of the halide, a gradual shift in the position of the main absorption peak can be observed, indicating that the tunable bandgap can be finely controlled by designing halide compositions reasonably. Yu et al.^[39] systematically studied the underlying mechanism of how halogen anions affect the optical and electrical properties of 2DRP perovskite based on $\text{PEA}_2\text{MA}_3\text{Pb}_4\text{I}_{11}\text{X}_2$ ($\text{X} = \text{I}, \text{Br}$ or Cl , $n = 4$). As shown in Figure 4f, the PL absorption peak of on $\text{PEA}_2\text{MA}_3\text{Pb}_4\text{I}_{11}\text{Br}_2$ film shows a blue shift, indicating the formation of low dimensional RP phase perovskites with wide bandgap. Due to poor carrier transport of low dimensional RP perovskites, this may impair the electrical properties of $\text{PEA}_2\text{MA}_3\text{Pb}_4\text{I}_{11}\text{Br}_2$ films. However, compared with

$\text{PEA}_2\text{MA}_3\text{Pb}_4\text{I}_{11}\text{Br}_2$ film, $\text{PEA}_2\text{MA}_3\text{Pb}_4\text{I}_{11}\text{Cl}_2$ film exhibits a red-shift with the lowest bandgap among $\text{PEA}_2\text{MA}_3\text{Pb}_4\text{I}_{11}\text{X}_2$. Notably, the chloride ions in $\text{PEA}_2\text{MA}_3\text{Pb}_4\text{I}_{11}\text{Cl}_2$ perovskite promote a vertically crystal orientation and thus improve the carrier transmission pathways in perovskite film.

Nevertheless, current reports on halogen anion modulation of the bandgap still focus on 3D perovskites,^[58] and the fine modulation of 2DRP perovskites needs further investigation.

4. Efficient Charge Carrier Transport in 2DRP Perovskite Thin Film

Perovskites with narrow bandgap can absorb more incident light and provide many carriers. How to achieve efficient carrier transport is a critical problem. Carrier transport in 2DRP perovskite films is affected by many factors. On the one hand, the phase orientation and phase distribution can affect the charge transport due to the quantum well. On the other hand, the chemical interaction between amine molecules and inorganic layers in 2D perovskites affects the charge transfer. Most 2D perovskites are prepared by the solution method, and the in-crystal defects by this method hinder the charge transfer process. Reasonable perovskite structure regulation to improve crystal crystallinity and prolong the hot carrier process effectively promotes charge carrier transport.

4.1. Perovskite Crystal Orientation

As shown in Figure 5a, carriers can only be transported along the inorganic octahedral layer due to the insulation of bulk amine molecules in 2DRP crystals. To realize the efficient carrier transporting from the perovskite layer to the transport layer, perovskite crystal orientation must grow along the direction perpendicular to the substrate.^[59] The study by Cao et al.^[47] has already pointed out, 2D materials grow in a parallel substrate direction when $n = 1$. When the n value increases, the crystal begins to grow in the direction outside the plane due to the growth competition between A' cation and A cation (Figure 5b). And as mentioned before, the exciton binding energy within the $n = 1$ perovskite phase is large, the bandgap is quite wide, and charge carrier transport is very difficult, making it unsuitable for use as an active layer material in photovoltaic devices. Therefore, most current crystal orientation studies focus on 2DRP perovskite phases with n values from 3 to 5.

In order to obtain vertical-oriented perovskite films, Tsai and co-workers^[16] prepared $(\text{BA})_2(\text{MA})_{n-1}\text{Pb}_n\text{I}_{3n+1}$ perovskites with different n values ($n = 3, 4$) by hot casting technique. The hot-cast films were found to have a denser and more uniform morphology and more obvious crystal orientation. As shown in Figure 5c, the crystal domains were oriented with their (101) planes parallel to the substrate surface and show 2D in-plane orientational randomness. High-quality films ultimately contributed to improved carrier mobility and reduced defect density within the device. Chen and co-workers^[60] improved the spin coating process by first crystallizing perovskite films on a hotplate at elevated temperature with minimal spin-induced convection and then rapidly annealing. $\text{BA}_2\text{MA}_3\text{Pb}_4\text{I}_{13}$ perovskite films with

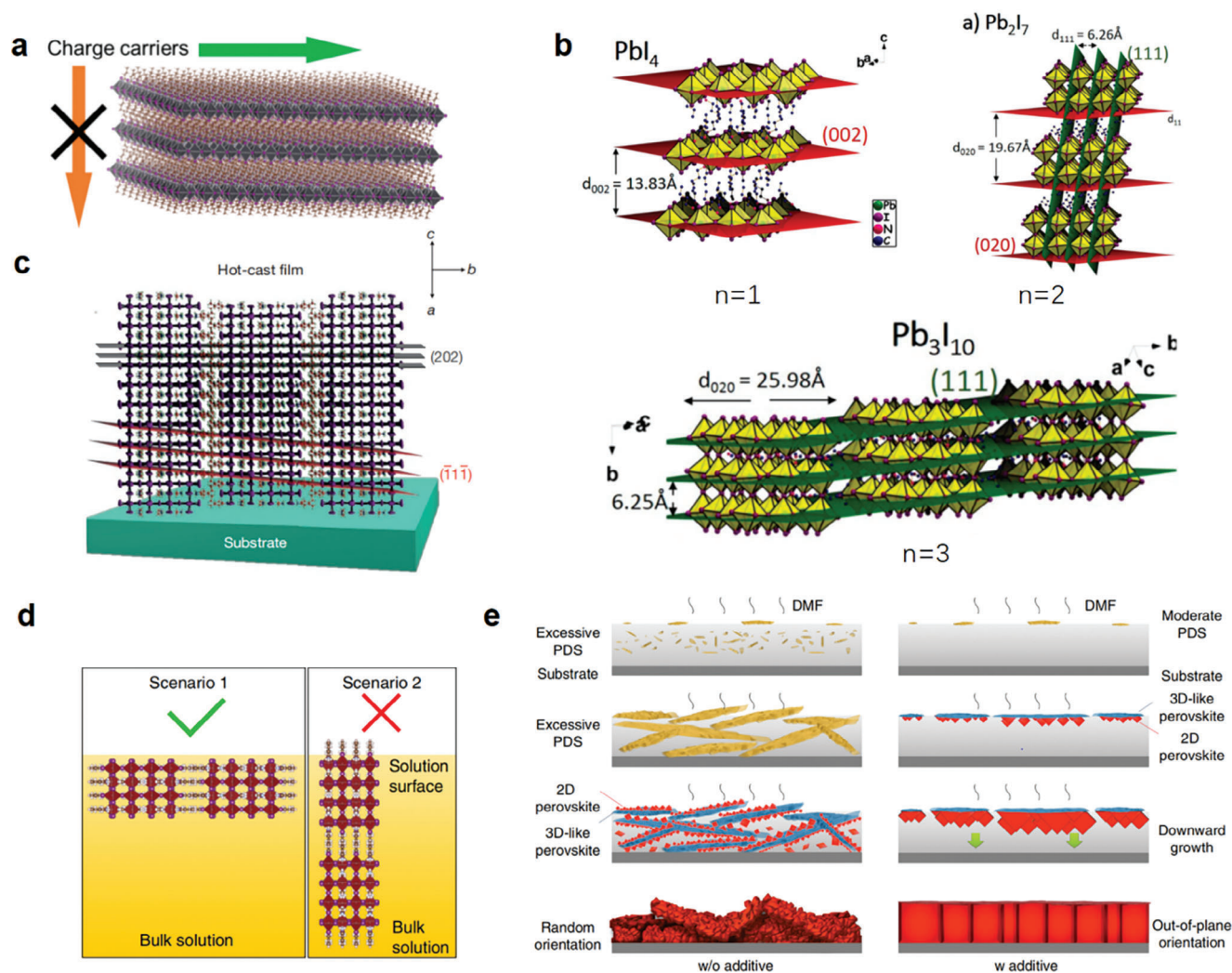


Figure 5. a) Schematics of charge transport anisotropy in a 2D perovskite thin film. In 2D perovskite, the organic layer is charge insulating, making it difficult for charge carriers to flow across layers. It is more favorable for charge carriers to flow along the inorganic plane direction. Reproduced with permission.^[59] Copyright 2019, AIP Publishing. b) The illustration of the diffraction planes of thin films versus bulk materials of BA_2PbI_4 , $(\text{BA})_2(\text{MA})\text{Pb}_2\text{I}_7$, and $(\text{BA})_2(\text{MA})_2\text{Pb}_3\text{I}_{10}$. Reproduced with permission.^[47] Copyright 2015, American Chemical Society. c) Schematic representation of the (101) orientation, along with the (111) and (020) planes of a 2D perovskite crystal, consistent with the GIWAXS data. Reproduced with permission.^[16] Copyright 2016, Springer Nature. d) Illustration of an orthorhombic (101) vertically oriented 2D perovskite structure, with (020) planes parallel to the substrate and schematic of preferred orientation at the liquid–air interface. Reproduced with permission.^[60] Copyright 2018, Springer Nature. e) Illustration of the 3D-like perovskites defined nucleation and growth of RP crystals from precursor solution without or with excessive AX additives. Reproduced with permission.^[61] Copyright 2020, Springer Nature.

high crystal orientation were prepared. They found that the nucleation of $\text{BA}_2\text{MA}_3\text{Pb}_4\text{I}_{13}$ is non-uniform, independent of the substrate, and occurs at the liquid–gas interface. As shown in Figure 5d, the initial orientation of the nucleus is the vertical configuration. They investigated the effects of different crystal growth orientations on carrier transport and device performance in 2D perovskite. It has been shown that, compared with partial random oriented films, the pre-crystallization annealed films with strong vertical orientation have higher short circuit current density and lower series resistance due to the smoother carrier transport channel.

In addition to adjusting the spin coating process, additive engineering is an effective way to control perovskite crystal ori-

entation. Zhang and co-workers^[62] revealed that 2D perovskite crystals $(\text{BA})_2(\text{MA})_{n-1}\text{Pb}_n\text{I}_{3n+1}$ ($n = 3, 4$) could be regulated by SCN^- -based additives because SCN^- has an ionic radius similar to that of I^- and the lone pair electrons from the S and N atoms can strongly interact with Pb^{2+} , possibly regulating the crystal crystallization process. Adding NH_4SCN reduced the energy level of $(\text{BA})_2(\text{MA})_2\text{Pb}_3\text{I}_{10}$ perovskite by 0.2 eV, and the device energy level matching was better and more conducive to charge transport. NH_4SCN greatly induced the vertical orientation of perovskite film and significantly improved the crystal quality. The PL lifetime of the perovskite films before and after the addition of NH_4SCN increased from 5.65 to 11.30 ns, which further verified this conclusion. Li and co-workers^[63] adopted Cl^- as an

additive to assist perovskite crystal growth. It was found that 2D perovskites have good vertical orientation with low Cl^- concentration. When the concentration of Cl^- increases, vertical and horizontal growth will occur in the crystal. Excessive Cl^- additive is not conducive to crystal orientation regulation, which may be related to the change in crystal formation energy. Wang and co-workers^[61] systematically analyzed the formation process of $(\text{BA})_2(\text{MA})_3\text{Pb}_4\text{I}_{13}$ layered perovskites and elucidated the mechanism by AX salts ($\text{A} = \text{NH}_4^+$ or MA^+ ; $\text{X} = \text{Cl}^-$, Br^- , or I^-) additives drive crystal vertical orientation (Figure 5e). The orientation is not directly determined by the preferential formation of RP crystals at the liquid–gas interface. However, it is induced by one-dimensional PbI_2 -DMF-contained solvate phases (PDS) before perovskite formation. Adding NH_4Cl enhances the solubility of the PbI_6 octahedron and inhibits the nucleation of PDS in the solution. On the contrary, PDS may be precipitated on the solution's surface due to the solvent's rapid evaporation. The formed PDS are conducive to the downward growth of RP phase perovskite and finally obtain vertical orientation. The work provides general guidance for the total regulation of crystal nucleation in different solution manufacturing processes. The additives also apply to the antisolvent. Lai and co-workers^[52] prepared high-quality FA-based perovskite films with 4-(trifluoromethyl) benzylamine-ammonium iodide (tFM-PMIAI) assisted crystal growth (OACG) technique. The tFM-PMIAI was dissolved in isopropyl alcohol (IPA) antisolvent to induce crystal growth and orientation. There is no (0k0) peak in the XRD pattern of the OACG-treated film, indicating that the crystal is mainly vertically oriented, and the degree of vertical orientation can be inferred from the peak strength. The optimized film strongly promoted charge extraction and transfer within the device, and the calculated electron and hole mobilities were 3.76×10^{-3} and $1.77 \times 10^{-3} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, respectively, which are both improved to different degrees compared to the untreated film. The device obtained an efficiency of 19.06% and significantly realized enhanced stability.

Different solvents specifically influence the crystal orientation of 2DRP phase perovskite. Cao and co-workers^[64] found that $(\text{BA})_2(\text{MA})_2\text{Sn}_3\text{I}_{10}$ perovskites showed crystal growth direction parallel to the substrate in dimethyl sulfoxide (DMSO) and vertically oriented in *N,N*-dimethylformamide (DMF). This might be related to the surface wettability of the solvent and transport layer and the formation of the intermediate complex $\text{SnI}_2 \cdot 3\text{DMSO}$ between DMSO and Sn^{2+} . Chen and co-workers^[65] proposed a crystallization mechanism for 2D perovskite. High supersaturation promotes nucleation, and low supersaturation promotes growth. They used three solvents, gamma-butyrolactone (GBL, $V_{\text{evap}} = 0.03$), dimethylacetamide (DMAc, $V_{\text{evap}} = 0.17$) and 2-methoxy ethanol (2ME, $V_{\text{evap}} = 0.5$), with different solvent evaporation rates (V_{evap} , evaporation rate of *n*-butylacetate reference is set as 1) to study the crystallization mechanism. It was found that the slower the solvent evaporation rate was, the more favorable vertical orientation preference was. Different solvents will have different solubility to the precursor, and different crystallization dynamics and processing conditions will affect different solvents. Zhang and co-workers^[66] proposed a binary solvent engineering to address the poor regulation effect of a single solvent. The highly vertically oriented 2D $(\text{BA})_2(\text{MA})_3\text{Pb}_4\text{I}_{13}$ perovskite films were prepared using DMF/DMSO as a mixed solvent. Moreover, the formation process was speculated. In the first

step, heterogeneous nucleation occurs at the gas–liquid interface, forming a thin, flaky cap layer on the surface. In a single DMF solvent, solvent volatilization is faster, resulting in uniform nucleation under the sheet cap layer and random crystal orientation. In mixed solvents, adding DMSO will form intermediate product BAI/MAI/ PbI_2 /DMSO, delaying solvent volatilization. The cap layer has enough time to act as a seed to induce the downward growth of the crystals.

It is clear that the key to obtaining vertical orientation is the precise regulation of the nucleation and growth process of perovskite crystals, regardless of the optimization of the deposition process or the choice of additives and solvents. It would be extremely beneficial if certain measures could be taken to lower the nucleation barrier of the crystals in order to increase the number of favorably oriented nuclei at the initial time and to retard the subsequent crystallization process. A large number of research works have proved that the formation of homogeneous and stable intermediates is the key to delay crystallization, and the introduction of strongly interacting ions, organic ligands, or molten salts of organic ligands into the precursors is undoubtedly a worthy direction to be considered in the future to improve the degree of vertical orientation of crystals.

4.2. Perovskite Phase Distribution

The stoichiometric ratio of each component in the precursor solution generally determines the *n* value of perovskites. As $n \geq 2$, various *n*-value perovskite phases tend to grow due to their self-assembly effect. The chaotic multiphase distribution hinders carrier transport and energy transfer, leading to the exacerbation of non-radiative complexation due to the aggregation of carriers in the low bandgap, and the degradation of device performance due to the difficulty in achieving efficient carrier extraction.^[67] Adjustment of the phase distribution of perovskites is of vital importance.

The preparation of pure phase perovskite is the most direct way to regulate current, although there are few reports on pure phase 2D perovskite. Liang and co-workers^[68] first reported the preparation of phase-pure perovskites with different quantum well widths. They introduced a molten salt spacer *n*-butylamine acetate (BAAC) to replace the traditional halide spacers *n*-butylamine iodide (BAI). Through strong ionic coordination between BAAC and perovskite skeleton, intermediate phase compounds with uniform distribution were gelled in the precursor solution. By ejecting the coordinated molten salt and evaporating the by-product methylammonium acetate (MAAC), homogeneous and fine intermediate phases can be crystallized into phase-pure quantum well (QW) films with micro-scale vertically aligned grains. Femtosecond transient absorption (TA) spectra show that all perovskite films prepared using BAAC are dominated by a single photobleaching (PB) peak, and the photobleaching is caused by cooled carriers in band edge states, suggesting that excited hot carriers were injected into a perovskite quantum well of comparable thickness, and then cooled down to the band-edge state afterward to produce consistent PB signals, which verifies the successful preparation of phase-pure QW films. The PCE of pure phase QW PSC is higher than that of multiple quantum

wells (MQW) PSC, reaching 16.25%, due to the significant increase in open-circuit voltage (V_{OC}) and fill factor (FF).

Most research focuses on charge carrier transport between mixed phases. Building on the study by Liu et al.^[27] Lin and colleagues^[69] investigated layered perovskites' phase separation mechanism and energy transfer process. Firstly, they prepared a compact and homogeneous layered perovskite film without pinholes by hot casting method and then characterized and analyzed its vertical and lateral phase segregation. In the vertical direction, the perovskite phase with small- n tended to stay at the bottom, while the layered phase with large- n or 3D-like perovskite phase tended to stay at the top. In the horizontal direction, the distribution of perovskite phases is also uneven. Layered perovskite is surrounded by 3D-like perovskite, as shown in **Figure 6a** (the dark area corresponds to the 3D-like perovskite phase, and the bright area corresponds to the layered perovskite phase). The 3D perovskite crystals tend to precipitate first in the process of growth. With the increase of precipitation, the ratio of BA to MA in the residual solution increases, and the BA-rich solution promotes the formation of layered perovskite. At this time, the BA/MA ratio gradually decreases, and the MA-rich solution again promotes the nucleation growth of the 3D crystal. The cambium layered perovskite is embedded into the 3D perovskite-like network, and with the increasing BA concentration, it is easier to form layered perovskite with a small n value as the lower the solution. Through this model, the working mechanism of the corresponding devices can be explained, and the photon absorbed by the layered perovskite phase can contribute photocurrent through the process of energy transfer from the more extensive bandgap layers to the smaller bandgap 3D-like phase, and the carrier transport inside perovskite mainly depends on the percolating network formed by the 3D phase (**Figure 6b**). Gradient phase distribution is beneficial to band alignment and carrier transport. Wang et al.^[70] used a modified constant thermal-annealing spin-coating (CTAS) process based on hot-casting to regulate the crystal growth process of 2D perovskite films by adding DMSO solvent to the precursor solution. They demonstrated that a refined band alignment strategy can effectively manipulate the phase at the bottom of perovskite to obtain a gradient phase distribution and vertically grown thin film structure, as shown in **Figure 6c**. Through space-charge-limited current (SCLC) measurements, they further determined the carrier transfer kinetics of the devices with different DMSO contents. The calculated electron mobility of $1.16 \times 10^{-3} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ in the DMSO 0.25 film was higher than that of $3.50 \times 10^{-4} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ in the DMSO 0 film, indicating improved charge extraction and transport in the perovskite layer, and the corresponding device ultimately achieved a PCE of 17.66%. Interestingly, after comparing MA-based and FA-based 2D perovskites films, Shao et al.^[54] found that, in contrast to MA-based perovskites that exhibit vertically oriented gradient phase distribution, pure FA-based perovskites have a large number of 3D-like phases distributed throughout the entire thickness of the film, forming a charge transport network in which the 3D-like phases are interspersed with the low-dimensional phases (**Figure 6d**). And the (TA) results confirmed that this special phase distribution favors ultrafast energy/charge transfer from 2D phases to 3D-like phases, and the measured carrier diffusion length reaches $1.85 \mu\text{m}$. Then they speculated that the mixed-phase morphology significantly enhances the light absorption

in the near-infrared region, which promotes charge dissociation and transport, resulting in a large enhancement of the photocurrent. Ultimately, the photovoltaic device based on pure FA-based thin films obtained a recorded efficiency of 21.07% with excellent thermal and environmental stability.

Adjusting the gradient phase distribution cannot wholly improve the hole transport because there are still low- n phases ($n \leq 2$) at the bottom of the perovskite film near the interface of the hole transport layer. The low- n phases with deep valence bands can generate a considerable hole extraction barrier, limiting the detrapping and transmission of charges and leading to significant carrier recombination.^[70] It is necessary to suppress the existence of low- n phases. Sidhik et al.^[71] designed a "phase-selective" method to prepare 2D perovskite films with uniform phase distribution by slowly crystallizing the perovskite single-crystal powders with a desired n -value from the precursor material to obtain the parent crystals, which were then dissolved in an optimized solvent and used for spin coating and annealing with the resulting solution (**Figure 6e**). In situ X-ray diffraction and absorbance tests show that the phase-selective synthesis method bypasses the intermediate phase so that the desired n -value phase is the primary phase at the time of initial nucleation. The perovskite films containing only $n = 3$ and $n = 4$ phases were prepared. The $\text{BA}_2\text{MA}_3\text{Pb}_4\text{I}_{13}$ device achieved a high efficiency of 17.1%, a high open-circuit voltage of 1.20 V, and a high fill factor of 81.1%. Wang et al.^[72] introduced trimethyl sulfides (TMS) into perovskite precursor solution for intermediates evolution and film quality control. The introduction of TMS inhibits the formation of undesired $(\text{PbI}_2)_2 \cdot \text{S}_2$ intermediate and promotes the formation of structured fibrous intermediates. Both steady-state PL spectroscopy and TA spectroscopy show that the low- n phase is inhibited in the films containing TMS compared to the control films, eliminating charge transport barriers in the perovskite films and reducing nonradiative recombination. In addition, the results of the transient photocurrent (TPC) measurement showed that the charge extraction time is reduced from 85.51 ns for the control device to 59.44 ns for the w/ TMSI device and 36.59 ns for the w/ TMSBr device, which suggested that the improvement in charge extraction is the key to the device performance improvement. The obtained photovoltaic devices realized an efficiency of up to 19.08%. In addition, vacuum poling treatment^[73] and other methods are also used to regulate the uniform phase distribution of high n values to improve current transport.

It can be easily seen from the above discussion that adjusting the preparation process of 2DRP perovskite properly is conducive for the improvement of the phase orientation and phase distribution. Therefore, **Table 1** summarizes the preparation characteristics of the perovskite and the corresponding film improvement for further clarification.

4.3. Organic Spacer Cations Regulation

Optimizing organic spacer cations (OSCs) is one of the common ways to improve device performance. The choice of OSC is directly related to the alignment of the perovskite layers. Good layer alignment affects the interactions of the inorganic interlayer orbitals, which in turn improves charge transport, reduces

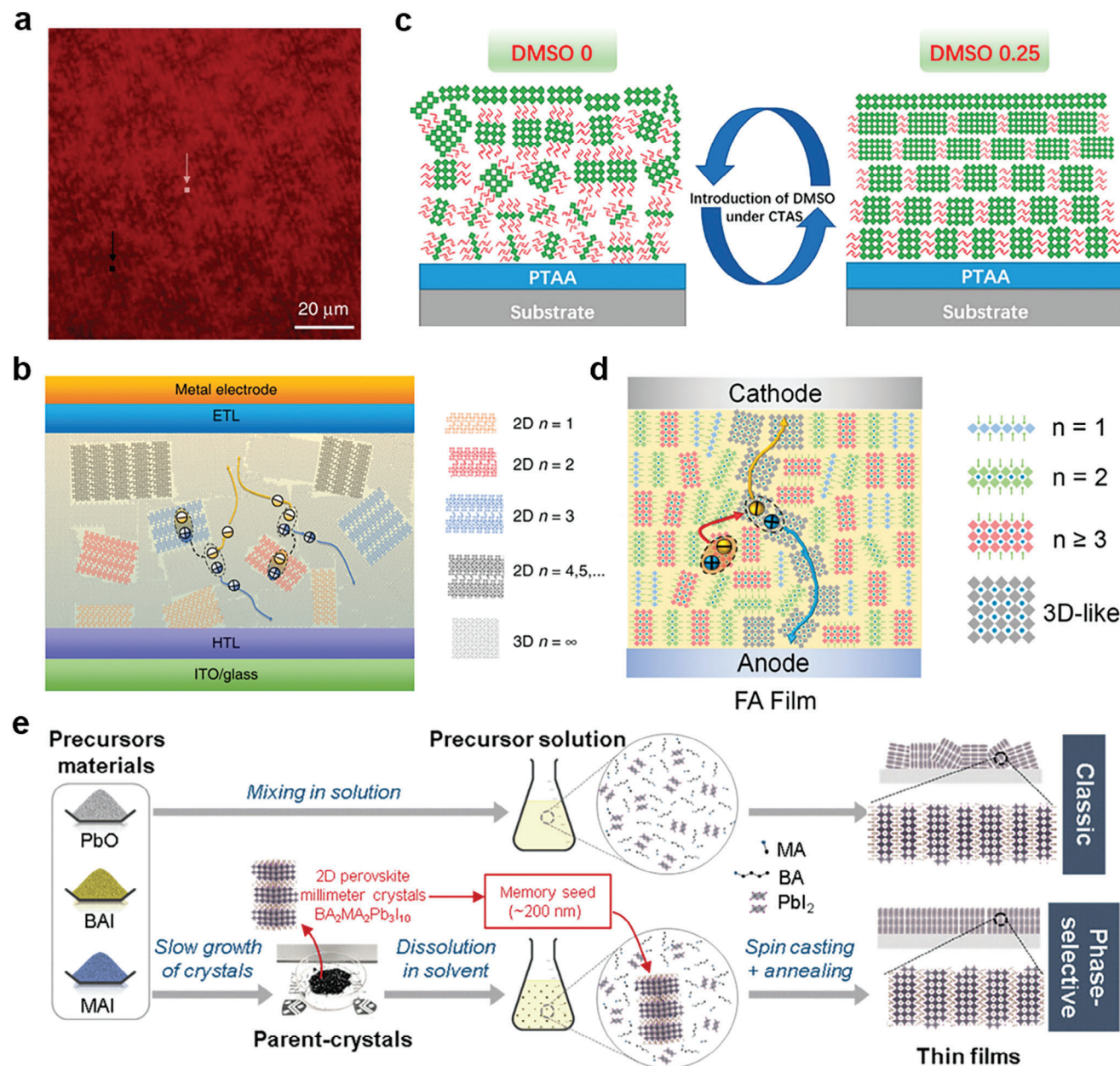


Figure 6. a) Optical mapping of the layered perovskite $(\text{BA})_2\text{MA}_3\text{Pb}_4\text{I}_{13}$ films using an optical microscope in transmission mode. Typical bright and dark areas are labeled with pink and black squares. Reproduced with permission.^[69] Copyright 2019, Springer Nature. b) Device operation mechanism model. (The device structure is indium tin oxide (ITO)/hole transport layer (HTL)/layered perovskite film/electron transport layer (ETL)/metal electrode). Reproduced with permission.^[69] Copyright 2019, Springer Nature. c) Schematics of the crystalline structures of the DMSO 0 and DMSO 0.25 films. (The volume ratios varied from 0% to 50%, denoted as DMSO 0 and DMSO 0.50.) Reproduced with permission.^[70] Copyright 2022, American Chemical Society. d) The schematic diagrams of the crystal orientation and phase distribution of FA-based 2D perovskites. Reproduced with permission.^[54] Copyright 2021, Wiley-VCH. e) Sketch of the phase-selective synthesis method from this work and the classic synthesis method already reported in the literature. Reproduced with permission.^[71] Copyright 2021, Wiley-VCH.

defect density and extends carrier lifetime. BA^+ and PEA^+ cations are the earliest and most representative OSCs. Cao et al.^[47] introduced $n\text{-BA}^+$ into the perovskite structure, and a perovskite film with ultrasmooth and high-surface coverage was prepared due to the steric effects of large cations. The addition of $n\text{-BA}^+$ improves the stability of perovskite, but it limits the current transport in perovskite film. Chen and co-workers^[84] designed

a spacer cation iso- BA^+ with a short branched chain and prepared $(\text{iso-BA})_2(\text{MA})_3\text{Pb}_4\text{I}_{13}$ thin films. Compared with the traditional linear structure, $(\text{iso-BA})_2(\text{MA})_3\text{Pb}_4\text{I}_{13}$ perovskite films show enhanced out-of-plane orientation, optical absorption, and crystallinity. Recently, a systematic study of two isomeric ligands, $n\text{-BA}^+$ and iso- BA^+ , from precursors to devices was carried out by Qin et al.^[85] It is shown that iso- BA^+ molecular clusters are more

Table 1. Summary of partial fabrication parameters and respective improvement of RP perovskite films.

2DRP perovskite	Characteristics of perovskite film spinning coating process				Annealing	Improvement	Ref.
	Hot-casting	Solvent	Antisolvent	Additive			
(BA) ₂ (MA) ₃ Pb ₄ I ₁₃	150 °C	DMF	Without	Without	Without	Phase orientation	[16]
(BA) ₂ (MA) _{n-1} Pb _n I _{3n+1}	Without	DMF	Without	NH ₄ SCN	100 °C	Phase orientation	[62]
(PTA) ₂ (MA) ₃ Pb ₄ I ₁₃	Without	DMSO	Without	Cl ⁻	100 °C	Phase orientation	[63]
(BA) ₂ MA ₄ Pb ₅ I ₁₆	70 °C	DMF, DMSO	Without	H ₂ O	100 °C	Gradient phase, Phase orientation	[74]
(ThMA) ₂ (FA) _{n-1} Pb _n I _{3n+1}	Without	DMF	tFM-PMAl	Without	100 °C	Phase orientation	[52]
(BA) ₂ (MA) _{n-1} Sn _n I _{3n+1}	120 °C	DMF	without	Without	75 °C	Phase orientation	[64]
(BA) ₂ (MA) ₃ Pb ₄ I ₁₃	120 °C	DMF, DMSO	Without	Without	100 °C	Phase orientation	[66]
BA ₂ MA _{n-1} Pb _n I _{3n+1}	100 °C	DMF, DMSO	Without	BAAc	100 °C	Pure phase	[68]
BA ₂ MA ₄ Pb ₅ I ₁₆	63 °C	DMF, DMSO	Without	MACl	70 °C	Gradient phase, Phase orientation	[70]
BA ₂ MA ₃ Pb ₄ I ₁₃	Without	DMF	Without	MACl	100 °C	Phase purity	[71]
AA ₂ MA _{n-1} Pb _n I _{3n+1}	65 °C	DMF, DMSO	Without	TMS	100 °C	Phase orientation, Phase purity	[72]
PEA ₂ MA _{n-1} Pb _n I _{3n+1}	100 °C	DMF, DMSO	Without	PbCl ₂	100 °C	Gradient phase	[73]
GA _{0.2} BA _{1.8} MA ₅ Pb ₆ I ₁₉	100 °C	DMF	Without	Formamide	100 °C	Phase orientation	[75]
(F-PEA) ₂ MA ₃ Pb ₄ I ₁₂	80 °C	DMF, DMSO	Without	MACl, PbCl ₂	100 °C	Phase orientation, Gradient phase	[76]
BA ₂ MA ₃ Pb ₄ I ₁₃	Without	DMF, DMSO(DMF, NMP)	Without	Without	100 °C	Phase orientation, Suppress small- <i>n</i> phase	[77]
(4FPEA) ₂ (MA) ₄ Pb ₅ I ₁₆	Without	DMF	Without	NH ₄ SCN	100 °C	Phase orientation, Gradient phase	[78]
(BA) ₂ (MA) ₃ Pb ₄ I ₁₃	100 °C	DMF, DMSO	Without	PbCl ₂	100 °C	Suppress small- <i>n</i> phase	[79]
(FPEA) ₂ (FA _{0.05} MA _{0.95}) ₄ Pb ₅ I ₁₆	Without	DMF	Without	NH ₄ SCN	100 °C	Phase orientation	[80]
FPEA ₂ MA ₃ Pb ₄ I ₁₃	100 °C	DMF, DMSO	Without	MACl	100 °C	Phase orientation, phase distribution	[81]
(PPA) ₂ (MA) ₂ Pb ₃ I ₁₀	Without	DMF, DMSO	Without	Pb(SCN) ₂	100 °C	Phase orientation	[82]
PEA ₂ Cs ₃ Pb ₄ I ₁₃	100 °C	DMF, DMSO	Without	MACl	100 °C	Gradient phase	[83]

spatially compact due to the stronger intermolecular binding energy and thus have stronger electrostatic and van der Waals binding compared to linear BA⁺ molecules. Ultimately, their study well explains the working mechanism of iso-BA⁺ ligands affecting two-dimensional perovskite crystallization and obtains photovoltaic devices with significantly higher short-circuit currents (≈20 mA cm⁻²).

Chao et al.^[86] used short-range bulkier organic ammonium, 1-amino-3-butene hydrochloride (BEACl), and 3-buten-1-amine hydrochloride (BYACl) to replace traditional BAI and improved charge transport by controlling the tunneling effect. The BEA- and BYA-based devices eventually realized high PCE of 16.1% and 15.1%, respectively. Enhancing the interaction between amine molecules by substituting certain atoms or groups on macromolecules positively affects segment self-assembly and charge transport within 2D layered perovskite.^[87] Ren et al.^[88] synthesized a bulky ammonium (the alkylammonium 2-(methylthio)ethylamine hydrochloride, MTEACl) by replacing a C atom in BA with an S atom. As shown in Figure 7a, the S–S interaction in the tail of MTEA can induce strong out-of-plane growth of 2DRP phase perovskite film, favorable to charge transport, charge separation, low trap density, and thermal stability. The carrier mobility in MTEA-based films was substantially increased compared to BA-based films, and the corresponding devices obtained a PCE of up to 18.06%.

PEA is commonly applied to 2DRP perovskites to improve perovskite film stability. Smith and colleagues^[35] prepared (PEA)₂(MA)₂[Pb₃I₁₀] and achieved a 4.73% PCE and a high open-circuit voltage of 1.18 V. Later, Kamminga et al.^[87] found that dif-

ferent alkyl chain lengths of phenyl ligands have specific effects on perovskite structure. For phenyl ligands with increasing carbon atom number (PMA, PEA, PPA, and PBA, where the phenyl (P) and ammonium (A) groups are separated by a methyl (M), ethyl (E), propyl (P), or butyl (B) group), the PbI₆-octahedron in inorganic layer gradually changes from corner-sharing to corner-sharing and face-sharing coexistence with the increase of alkyl chain length. The structural bandgap induced by the long chain is unfavorable to the charge transport. PPA and PBA are not entirely suitable as organic ligands in 2D perovskites. Xi et al.^[92] replaced one C–C single bond in PPA with a double bond. They found that the sp² hybridization double bond causes the spacer to produce a significant hydrogen contact with I⁻ in the inorganic octahedron, allowing the perovskite to grow preferentially along the vertical orientation. Moreover, the PPA-based perovskite cell achieved a high PCE of 14.76%. In addition to altering the alkyl group, the researchers considered substituting other atoms for the H in the benzene ring in PEA, such as halogen atoms F, Cl, Br, and O atoms.^[93,94] Zhang et al.^[95] designed a new organic spacer 4-fluorophenethylammonium (4F-PEA) by fluorine substitution on the para position. They found that the inorganic interlayer force is enhanced, and the charge transfer is more efficient due to the reduced average ring centroid-to-centroid distances in the organic layer. Compared with the original PEA-based films and devices, the carrier lifetime in the fluorinated films is extended by 50%, and the efficiency of the corresponding devices is improved by nearly 4%. It is worth mentioning that the recorded efficiency of 2DRP phase perovskite solar cells reported in the literature so far was also realized based on the 4F-PEA ligand.^[54]

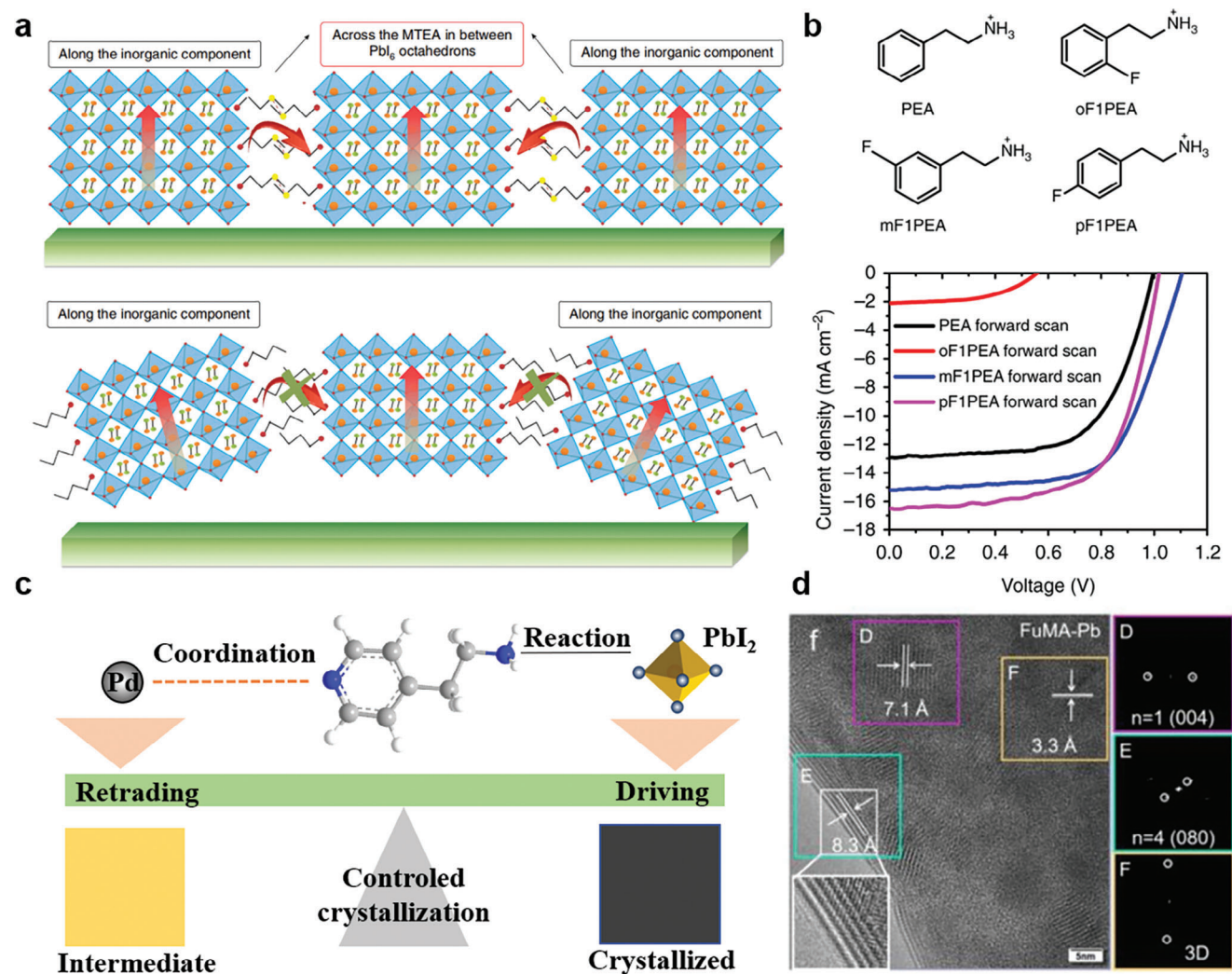


Figure 7. a) Schematics of MTEA and BA-based 2D perovskite film charge transport. Reproduced with permission.^[88] Copyright 2020, Springer Nature. b) 2D OIHPs with PEA fluorinated at different positions and their device characterization. Reproduced with permission.^[89] Copyright 2019, Springer Nature. c) The diagram of the 4-AEP control crystallization. Reproduced with permission.^[90] Copyright 2019, American Chemical Society. d) HRTEM images of the FuMA-Pb perovskite films. The right side shows fast Fourier transforms (FFT) images of the corresponding areas. Reproduced with permission.^[91] Copyright 2022, American Chemical Society.

As shown in Figure 7b, Hu et al.^[89] proposed that monofluorinating at different positions on the benzene ring has different effects on film properties. They named the alkyl ortho-, meso-, and para-fluorinated molecules as oF1PEA, mF1PEA, and pF1PEA, respectively, and analyzed the phase distribution, surface morphology, and single crystal structure. MF1PEA- and pF1PEA-based devices improve performance, while oF1PEA-based devices perform poorly. This work provides guidance for the design of materials with desirable functional device properties. Similarly, Yang et al.^[96] substituted Br for H in BA, named 3-bromobenzylammonium iodide (3BBAI), to prepare 2D PSCs. Vertically oriented high-quality films with unique structures (low n phase covered by high n phase) were obtained, and a peak PCE of 18.20% was realized.

In addition to the standard aliphatic spacers and aromatic spacers in the 2DRP phase perovskite, thiophene, pyridine, piper-

dine, and imidazole with unique functional groups have also been developed to improve the performance of PSCs. Lai et al.^[53] first developed 2-thiophenemethylammonium (ThMA^+) as a spacer cation in 2D perovskite. By MACl-assisted casting technology, perovskite thin films with nano-rod-like morphology were obtained. Due to the increase in crystal size and the enhancement of out-of-plane orientation, the corresponding perovskite devices achieved a PCE of 15.42%, the highest recorded at that time. Later, ThMA^+ was used to prepare pure FA-based perovskite devices and achieved more than 19% efficiency.^[52] Moreover, the FA-based thiophene derivative, 2-thiophenformamidinium (ThFA^+), was prepared to manufacture efficient and stable perovskite photovoltaic devices.^[97] Generally, suppose the hydrogen bond between the organic spacer layers and the inorganic slabs is too strong. In that case, the crystallization rate of perovskite is too fast, and the film's surface will be rough.^[98] Li and co-workers^[90]

Table 2. Parameters related to carrier transport of different 2DRP perovskites.

2DRP perovskite	τ [ns]	N_t^e [$\times 10^{15}$ cm $^{-3}$]	N_t^h [$\times 10^{15}$ cm $^{-3}$]	μ_e [$\times 10^{-3}$ cm 2 V $^{-1}$ s $^{-1}$]	μ_h [$\times 10^{-3}$ cm 2 V $^{-1}$ s $^{-1}$]	Ref.
(MTEA) $_2$ (MA) $_4$ Pb $_5$ I $_{16}$	122	2.22	5.18	13	17	[88]
(ThMA) $_2$ (FA) $_4$ Pb $_5$ I $_{16}$	4510	5.74	3.82	3.76	1.77	[52]
(NpMA) $_2$ (MA) $_3$ Pb $_4$ I $_{13}$	560	2.18	1.49	1.97	1.88	[99]
(BM) $_2$ (BA)MA $_3$ Pb $_4$ I $_{13}$	—	13	13	34	37	[100]
(ThFA) $_2$ (MA) $_2$ Pb $_3$ I $_{10}$	994.48	8.33	2.01	0.701	0.28	[97]
(PEA) $_2$ (MA) $_3$ Pb $_4$ I $_{13}$	6.10	—	4.52	—	7.76	[101]
(4F-PEA) $_2$ (FA) $_4$ Pb $_5$ I $_{16}$	986	3.41	3.74	1.3	1.1	[54]
FA $_2$ FA $_2$ Pb $_3$ I $_{10}$	297	7.21	—	139	—	[102]
(PA) $_2$ (MA) $_4$ Pb $_5$ I $_{16}$	35.1	6.45	—	3.04	—	[103]
(FuMA) $_2$ MA $_3$ Pb $_4$ I $_{13}$	18 030	7.68	19.9	2.57	0.256	[91]
(BA) $_2$ (FA) $_3$ Pb $_4$ I $_{13}$	144.3	6.90	—	—	—	[104]
(AA) $_2$ MA $_3$ Pb $_4$ I $_{13}$	140	3.96	1.95	2.50	0.271	[72]
(BThMA) $_2$ MA $_4$ Pb $_5$ I $_{16}$	495.0	4.63	9.56	20.4	3.82	[105]
(GABA) $_2$ MA $_4$ Pb $_5$ I $_{16}$	328	—	—	1610	—	[106]
(β -FPEA) $_2$ (FA) $_4$ Pb $_5$ I $_{16}$	—	5.59	7.57	7.98	4.28	[107]
PEA $_2$ MA $_4$ Pb $_5$ I $_{16}$	—	4.077	3.938	839.7	435.7	[108]
(HA) $_2$ FA $_3$ Pb $_4$ I $_{13}$	268.5	9.09	—	—	—	[109]
(TTFA) $_2$ MA $_4$ Pb $_5$ I $_{16}$	2030	13.1	19.9	4.70	0.29	[110]
((5FPTMA) $_{0.1}$ (PTMA) $_{0.9}$) $_2$ MA $_4$ Pb $_5$ I $_{16}$	2580	0.633	0.949	2.72	2.67	[111]
(3, 3-DFAz) $_2$ MA $_3$ Pb $_4$ I $_{13}$	511.07	7.68	2.56	5.12	1.41	[112]

reported that using 4-(aminoethyl)-pyridine (4-AEP) as an organic spacer cation can regulate the rate of perovskite crystallization. The aminoethyl cation in 4-AEP can react with PbI $_2$ to drive crystallization. Moreover, the N atom on the pyridyl unit can cooperate with Pb $^{2+}$ to delay crystallization and control the nucleation and growth process of perovskite crystals (Figure 7c). In this way, the high-quality and reproducible perovskite films can be obtained. The mechanism of furan as a ligand in 2DRP phase perovskites is very different from that of pyridine. Wang et al.^[91] compared furan-2-ylmethanaminium chloride (FuMACl) with an aromatic furan ring and (tetrahydrofuran-2-yl) methanaminium chloride (THFMACl) with an aliphatic THF ring. As shown in Figure 7d, it was found that the FuMA-based perovskite films with conjugated furan units exhibit apparent intermolecular π – π interaction and CH $\cdots\pi$ interaction, which is reflected by the change in crystal face spacing in the HRTEM images, leading to grain enlargement, better crystal orientation, and improvement of carrier lifetime extension. The obtained 2D perovskite films realized an average carrier lifetime of up to 18.03 μ s and a photovoltaic device with a PCE of 18.00% and FF of 81.95%. Based on the design of organic amine molecules and the optimization of film preparation means, the charge carrier transport properties within 2DRP perovskite films have been significantly improved in recent years, and Table 2 summarizes the relevant work of the last four years, with some of the parameters listed to help understanding.

From the above analysis, it can be seen that the organic ligands currently used in 2DRP perovskite solar cells are mainly of two types, aliphatic and aromatic. Among them, the aromatic ligands have significantly higher dielectric constants than the aliphatic ligands due to their conjugated π -electronic structures, which are more compatible with the dielectric properties between the inor-

ganic layers. Therefore, most of the 2DRP perovskite photovoltaic devices with better performance are realized based on aromatic ligands. As shown in Figure 8, it can be seen that thanks to the precise design of the amine molecules, the photovoltaic performance of 2DRP PSCs has been dramatically improved in just a decade. In particular, the PCE of 2D perovskite photovoltaic devices based on aromatic ligands has exceeded 20%, which is expected to realize a further breakthrough in high efficiency and stability.

Consequently, the later work should also focus more on aromatic ligands when designing organic amine ligands, and introduce heteroatoms and special chemical bonds on the basis of

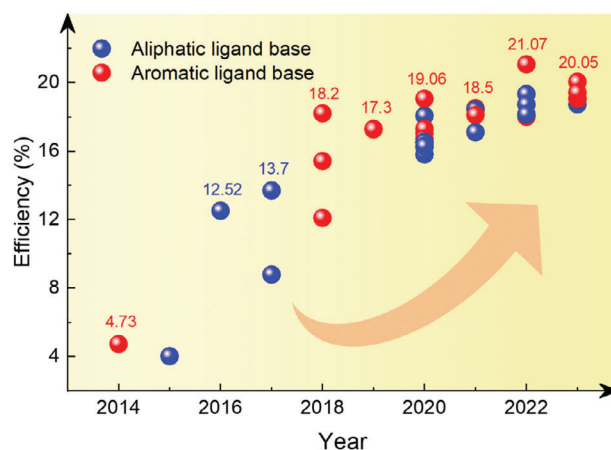


Figure 8. Summary of PCE evolution of 2DRP PSCs reported in the literature.

the existing ligands to enhance the intermolecular forces and the forces between organic molecules and inorganic octahedra, and then realize the modulation of the crystallization process and the improvement of photovoltaic performance.

5. Efficient Charge Carrier Transport in 2DRP Perovskite Device

Perovskite materials as the active layer are far from enough to ensure the integral carrier transport inside the PSCs. In order to guarantee that electrons and holes can enter the external circuit along their respective transmission paths, a particular energy level gradient and internal electric field between the various materials inside the device are necessary. So far, researchers have mainly focused on two aspects to improve the overall carrier transport level of the 2DRP PSCs. One is the appropriate selection of electrode and carrier transport layers to ensure the level matching, and the other is the modification at different interfaces to help carriers pass through the functional layers. Further performance enhancement of 2DRP perovskite photovoltaic devices is expected to be achieved by optimizing the energy levels and properties between different interfaces. **Table 3** summarizes the photovoltaic performance of different amine molecules and different device structures in recent years.

5.1. Energy Level Matching

The 2DRP perovskites with multiple quantum well structures have relatively complex energy level arrangements, so it is necessary to select the appropriate charge transport layer according to the multiphase distribution law of perovskites and take certain measures to optimize the energy levels at different interfaces. The alignment without energy level barriers is conducive to achieving higher charge extraction efficiency and lower energy loss to facilitate charge carrier transport.

5.1.1. *n-i-p* Structured Devices

Theoretically the gradient distribution of 2D perovskite phases with different n values can be regarded as a type II quantum well. After the dissociation of photogenerated excitons, the electrons transferred along the conduction band tend to move from the low- n value to the high- n value quantum well, whereas the holes are exactly the opposite (Figure 9a).^[117] Therefore, in conventional *n-i-p* devices, the bottom high- n phase favors the injection/collection of electrons, and the top low- n phase favors the injection/collection of holes.^[118]

In order to achieve effective energy level matching, researchers often need to use antisolvent to change the growth direction of the perovskite film to obtain a vertical distribution that satisfies the conventional devices,^[120] and then choose a charge transport layer with a lower energy level match. Several transport layer materials and their energy levels currently commonly used in 2DRP perovskite devices are shown in Figure 9b. Among them, TiO_2 and SnO_2 are both more common electron transport layers and

have both achieved good results when applied to perovskite photovoltaics. Liu et al. prepared a dense TiO_2 layer by chemical bath deposition^[121] and obtained 2D perovskite solar cells with PCE up to 19.33%.^[102] Actually, SnO_2 has more spartan preparation conditions than TiO_2 and has been reported relatively more recently. Related studies have shown that the SnO_2 film as the electron transport layer in the 2D perovskite device shows good charge transfer capability and low defect-induced recombination rate,^[63] which facilitates the optoelectronic performance of 2DRP PSCs. The most commonly used hole-transporting materials in 2D perovskites are often small organic molecules such as Spiro-OMeTAD. Spiro-OMeTAD is widely used due to its suitable energy level distribution, and its molecular structure is shown in Figure 9c. However, there is a particular energy level mismatch between Spiro-OMeTAD and the low n -value phases ($n = 1, 2$) in perovskites, which makes it difficult to achieve efficient hole extraction. Therefore, in the research work on conventional *n-i-p* structures, the key point is adding additives to the precursor to suppress the presence of low- n phases and to obtain a phase arrangement with an n value concentrated at 3–5 to enhance the hole injection. Since MACl can form an intermediate state to slow down the crystallization rate, Zhou et al.^[119] added MACl to the perovskite precursor to obtain perovskite phases with narrow n -value distributions, verified by TA spectroscopy (Figure 9d). It can be seen that in the *n-i-p* structure, in order to promote the transport of holes, it is necessary to control the phase arrangement in the 2DRP perovskites. Moreover, it is necessary to appropriately increase the energy level of Spiro-OMeTAD by a doping way. In fact, since the 2DRP perovskite itself tends to be arranged from small n at the bottom to large n at the top, only a tiny portion of the literatures uses Spiro-OMeTAD as the hole transport material,^[102,104,122] and the vast majority utilizes the *p-i-n* structure.

5.1.2. *p-i-n* Structured Devices

The ordered phase arrangement from small n at the bottom to large n at the top in 2DRP perovskites has been shown to be universal, making the *p-i-n* inverted structure as the better choice for 2D perovskite photovoltaics. A variety of charge carrier transport layer materials have been developed to match different energy levels of 2D perovskites for efficient charge carrier transport, and Figure 10a illustrates the energy level arrangement of several commonly used materials.

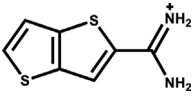
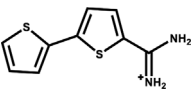
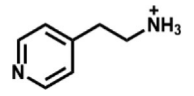
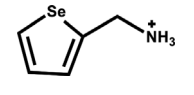
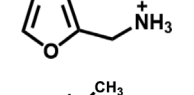
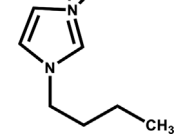
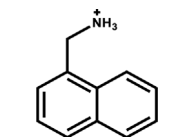
Fullerenes (C_{60}) and their derivatives are widely used as electron transport materials because they act as electron acceptors, forming donor–acceptor contact interfaces for charge separation. Among them, indene C60 bis-adduct (ICBA) is less used due to its higher energy level and particular energy level mismatches with the prevailing high- n phase in current 2DRP perovskites.^[6,6]-phenyl-C60-butyric acid methyl ester (PCBM) has a relatively low CB, which is more favorable for electron injection from most 2D perovskites. To further promote the charge carrier transport and electron extraction in 2DRP perovskite cells, a predeposition transport layer strategy was proposed by Fu et al.,^[116] where a thin layer of low-concentration PCBM was prepared on perovskite before the deposition of the electron transport layer (Figure 10b). It was shown that the conduction band

Table 3. Summary of 2DRP PSC photovoltaic parameters.

Amine molecule	Amine molecule structure	Device configuration	V_{OC} [V]	J_{SC} [mA cm ⁻²]	FF [%]	PCE [%]	Ref.
<i>n</i> -BA ⁺		FTO/SnO ₂ /(BA) ₂ (FA) ₃ Pb ₄ I ₁₃ /Spiro-OMeTAD/Au	1.062	21.62	78.96	18.14	[104]
iso-BA ⁺		ITO/PEDOT:PSS/(iso-BA) ₂ MA ₄ Pb ₅ I ₁₆ /PC61BM/BCP/Ag	≈1.0	≈20	–	≈15	[85]
BEA ⁺		ITO/PEDOT:PSS/(BEA) ₂ (MA) ₃ Pb ₄ I ₁₃ /PCBM/LiF/Al	1.01	20.63	78.0	16.1	[86]
BYA ⁺		ITO/PEDOT:PSS/(BYA) ₂ (MA) ₃ Pb ₄ I ₁₃ /PCBM/LiF/Al	1.01	19.53	76.4	15.1	[86]
MTEA ⁺		ITO/PEDOT:PSS/(MTEA) ₂ (MA) ₄ Pb ₅ I ₁₆ /PC61BM/Cr/Au	1.088	21.77	76.27	18.06	[88]
GABA ⁺		ITO/SnO ₂ /(GABA) ₂ MA ₄ Pb ₅ I ₁₆ /Spiro-OMeTAD/Au	1.11	22.16	76.16	18.73	[106]
AA ⁺		ITO/PTAA/AA ₂ MA ₃ Pb ₄ I ₁₃ /C ₆₀ /BCP/Ag	1.24	18.57	80.29	18.42	[113]
HA ⁺		FTO/SnO ₂ /(HA) ₂ FA ₃ Pb ₄ I ₁₃ -SENA/Spiro-OMeTAD/Au	1.09	23.60	77.86	20.03	[109]
PEA ⁺		ITO/2PACz/PEA ₂ MA ₄ Pb ₅ I ₁₆ /PC60BM/PEI/Ag	1.22	20.71	79.36	20.05	[108]
PMA ⁺		ITO/PEDOT:PSS/(PMA) ₂ (MA) ₂ Pb ₃ I ₁₀ /PCBM/BCP/Ag	1.064	18.10	70.50	13.58	[114]
PPA ⁺		ITO/PTAA/(PPA) ₂ (Cs _{0.05} (FA _{0.88} MA _{0.12}) _{0.95}) ₃ Pb ₄ (I _{0.88} Br _{0.12}) ₁₃ /C ₆₀ /BCP/Cu	1.14	18.72	66.19	14.16	[92]
4F-PEA ⁺		ITO/PTAA/(4F-PEA) ₂ (FA) ₄ Pb ₅ I ₁₆ /PCBM/BCP/Ag	1.18	22.45	79.53	21.07	[54]
β-FPEA ⁺		ITO/PTAA/(β-FPEA) ₂ (FA) ₄ Pb ₅ I ₁₆ /PC61BM/BCP/Ag	1.13	22.13	76.71	19.11	[107]
3FBA ⁺		ITO/PTAA/3FBAI-based PVK/PC61BM/BCP/Ag	1.223	20.98	78.41	20.12	[115]
3BBA ⁺		ITO/PTAA/3BBA ₂ MA _{n-1} Pb _n I _{3n+1} (3 < n < 4)/PC61BM/Cr/Au	1.23	18.22	81.2	18.20	[96]
ThMA ⁺		ITO/PEDOT:PSS/(ThMA) ₂ (FA) ₄ Pb ₅ I ₁₆ /PCBM/BCP/Ag	1.075	23.39	75.8	19.06	[52]
ThFA ⁺		ITO/PEDOT:PSS/(ThFA) ₂ MA ₂ Pb ₃ I ₁₀ /PCBM/BCP/Ag	1.05	20.17	79.0	16.72	[97]

(Continued)

Table 3. (Continued).

Amine molecule	Amine molecule structure	Device configuration	V_{oc} [V]	J_{sc} [mA cm ⁻²]	FF [%]	PCE [%]	Ref.
TTFA ⁺		ITO/PEDOT:PSS/(TTFA) ₂ MA ₄ Pb ₅ I ₁₆ /PC ₆₁ BM/BCP/Ag	1.12	22.16	78.21	19.41	[110]
BTFA ⁺		ITO/PEDOT:PSS/(BTFA) ₂ MA ₄ Pb ₅ I ₁₆ /PC ₆₁ BM/BCP/Ag	1.09	18.35	77.14	15.43	[110]
4-AEP ⁺		FTO/C60/(4-AEP) ₂ MA ₄ Pb ₅ I ₁₆ /Spiro-OMeTAD/Au	1.11	13.66	77	11.68	[90]
SeMA ⁺		ITO/PEDOT:PSS/(SeMA) ₂ FA ₄ Pb ₅ I ₁₆ /PDTL/PCBM/BCP/Ag	1.08	23.00	76.52	19.03	[116]
FuMA ⁺		ITO/PEDOT:PSS/(FuMA) ₂ (MA) ₃ Pb ₄ I ₁₃ /PCBM/BCP/Ag	1.054	20.84	81.95	18.00	[91]
BM ⁺		ITO/SnO ₂ /(BM)(BA)MA ₃ Pb ₄ I ₁₃ /Spiro-OMeTAD/Au	1.15	19.8	76.3	17.3	[100]
NpMA ⁺		ITO/PEDOT:PSS/(NpMA) ₂ (MA) ₃ Pb ₄ I ₁₃ /PCBM/BCP/Ag	1.24	20.89	66.35	17.25	[99]

energy level (E_{CB}) of the predeposition transport layer-treated perovskites was lowered. In contrast, the Fermi level (E_F) was shifted upward, yielding a thin film with more *n*-type characteristics and better matching with the energy levels (Figure 10c), and finally, achieving more efficient electron collection by the PCBM layer.

Comparatively, there is a broader choice of hole transport layers (HTL), divided into two main categories: organic materials and inorganic materials. However, it cannot be ignored that none of the existing hole-transporting materials exactly matches the low-*n* phase in 2D perovskites, and thus, a large amount of work has been focused on modifying the HTL to improve the energy level of the perovskite phase and facilitate the hole extraction. Tian et al.^[123] introduced a thin poly[(9,9-bis(3'-(*N,N*-dimethylamino)propyl)-2,7-fluorene)-alt-2,7-(9,9-dioctylfluorene)] (PFN) interface layer on poly(triarylamine) (PTAA), and a 2DRP perovskite film PEA₂MA_{3.5}FA_{0.5}Pb₅I₁₆ was obtained, in which the phase (*n* = 1) was significantly suppressed. The TRPL results show that the carrier lifetime of the films grown on PTAA/PFN is about twice as long as that of the pristine films (Figure 10d), suggesting that lowering the energy level barriers can promote the hole extraction and injection from the perovskite layer to the PTAA layer. Furthermore, doping HTLs with high work function materials can also effectively improve the energy level match between the HTL

and 2DRP perovskite layer.^[124] Inorganic HTLs such as NiO_x also have suitable valence band energy levels but are very few used in 2DRP perovskite solar cells.

Although the abovementioned materials have been developed and widely used, the researchers found that the thick HTL may decrease fill factor of PSCs, while too thin HTL would lead to direct contact between perovskite and electrodes.^[125] In fact, the thickness of HTL based on spin-coating or vacuum deposition is difficult to control, and thus the corresponding device shows poor repeatability. On this basis, self-assembled monolayers (SAMs) that can be chemically attached on a surface were developed and considered an effective alternative due to its forming extremely thin and stable layers.^[126,127] In 2019, Amran et al. combined carbazole bodies with phosphonic acid and designed two promising SAM binding groups, [2-(9H-carbazol-9-yl)ethyl]phosphonic acid (2PACz) and [2-(3,6-dimethoxy-9H-carbazol-9-yl) ethyl]phosphonic acid (MeO-2PACz),^[128] which made great progress of SAMs applied into perovskite solar cells. In 2020, the presence of [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz) attracted more attention in the field of PSCs.^[129] At present, SAMs have been reported to be applied to 2DRP perovskite optoelectronic devices to improve the energy band level.^[108] If the traditional HTL can be effectively combined with SAMs, it is believed that the charge carrier transport channel in 2DRP PSCs will be greatly dredged.

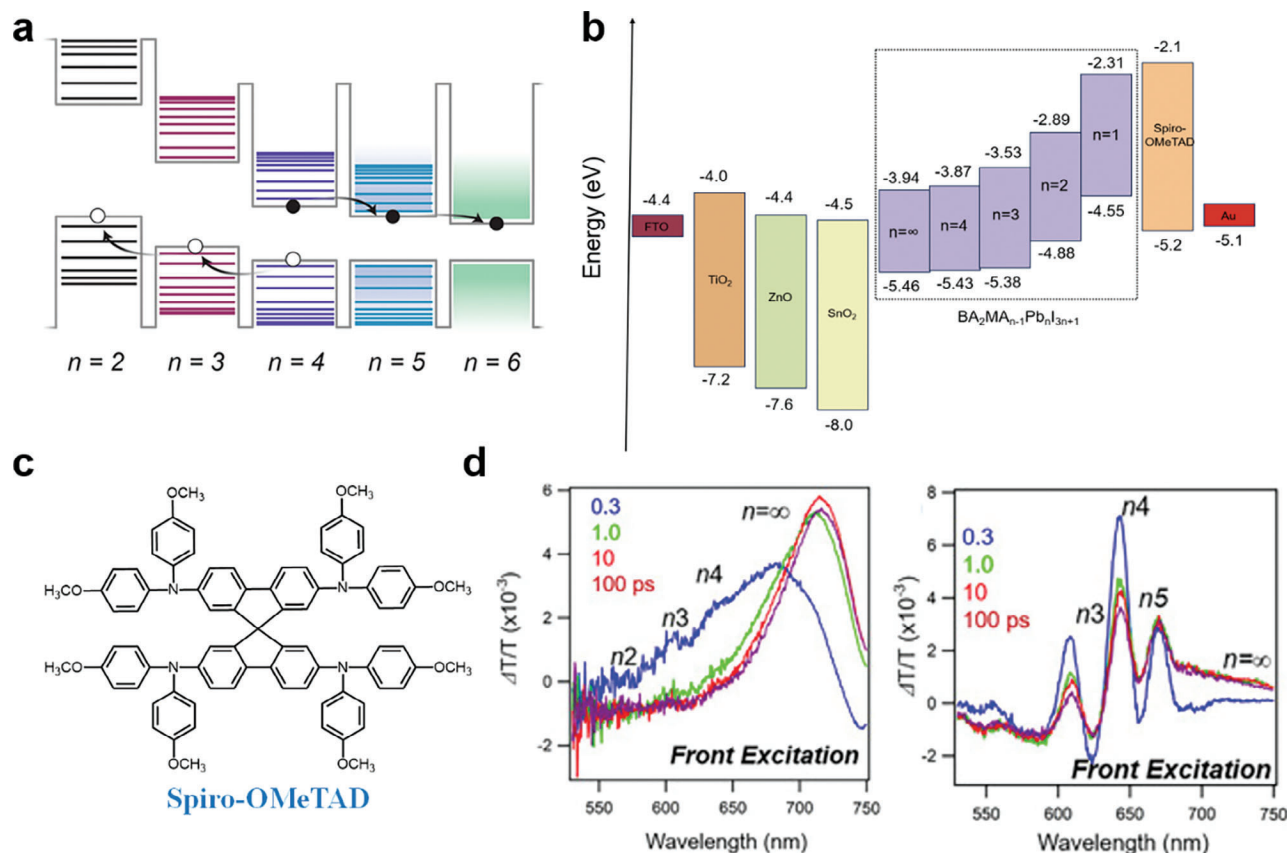


Figure 9. a) Schematic of the band structure for mixed low- n perovskite QWs. After photoexcitation and exciton dissociation, electrons will transfer from low- to high- n QWs, while the opposite occurs for holes owing to the type II band alignment. Reproduced with permission.^[117] Copyright 2018, American Chemical Society. b) Energy levels of different materials acting as inorganic ETM and organic HTM in n -i-p structured perovskite PSCs. c) Molecular structures of Spiro-OMeTAD. d) TA spectra at different time delays, kinetic traces and fitting at selected wavelength without/with addition of MACI at front excitation. Reproduced with permission.^[119] Copyright 2019, Wiley-VCH.

5.2. Interfacial Modification

The selection of ETL and HTL with appropriate energy levels is the premise to ensure carrier transport in perovskite devices. The modification of the ETL or HTL/perovskite layer interface is the focus of the research. The leveling of the interface continues to improve the quality of perovskite film. It also ensures effective covering and contact between layers, avoiding the increase of charge recombination and the significant decrease of V_{OC} and FF caused by direct contact between ETL and HTL.

5.2.1. Surface Modification

Perovskite films prepared by the solution method inevitably produce a large number of ionic defects and grain boundary defects during the formation process, and the deep energy level defects will act as nonradiative recombination centers to cause carrier recombination during device operation.^[130] Effective surface passivation is a robust way to solve this problem. In 2DRP perovskite devices, surface passivation reduces the defect density, promotes the 2D phase rearrangement, and obtains crystal orientation and arrangement favorable for charge transport. Huang et al.^[131] adopted the GABr treatment to regulate the secondary crystal-

lization in the surface layer of perovskite film $\text{GA}_2\text{MA}_4\text{Pb}_5\text{I}_{16}$. TA spectra show that the introduction of GA^+ changes the nucleation barriers of the different phases in the surface layer of perovskites, suppressing the reappearance of the low- n phase with larger exciton binding energy (Figure 11a), and the reduction of the low- n phase drives down the defect density, resulting in optimized films with long PL lifetime (Figure 11b). Due to the more uniform film distribution, the GABr-treated devices achieved efficient charge extraction/transport and obtained a PCE of up to 19.3%.^[131] Kim et al.^[108] used amphipathic n -tert-butyl- α -phenylnitron (PBN) with three functional groups (Figure 11c) to treat the 2D perovskite $\text{PEA}_2\text{MA}_4\text{Pb}_5\text{I}_{16}$ surface. Under the synergistic effect of different groups, PBN first induced vertical orientation of the surface layer of perovskites by dissolving them. Secondly, due to the passivation of defects, the recombination within the 2DRP perovskite film was minimized, and the PBN treatment induced an efficient energy transport from the small n -phase to the large n -phase (Figure 11d), leading to effective radiative recombination in the large n -phase and prolonging the carrier diffusion length.

Accordingly, to passivate multiple defects in 2DRP perovskites, molecules or ions with multiple functional groups can be selected, but the effect of the passivating material on the orientation and distribution of the 2D phase must be considered.

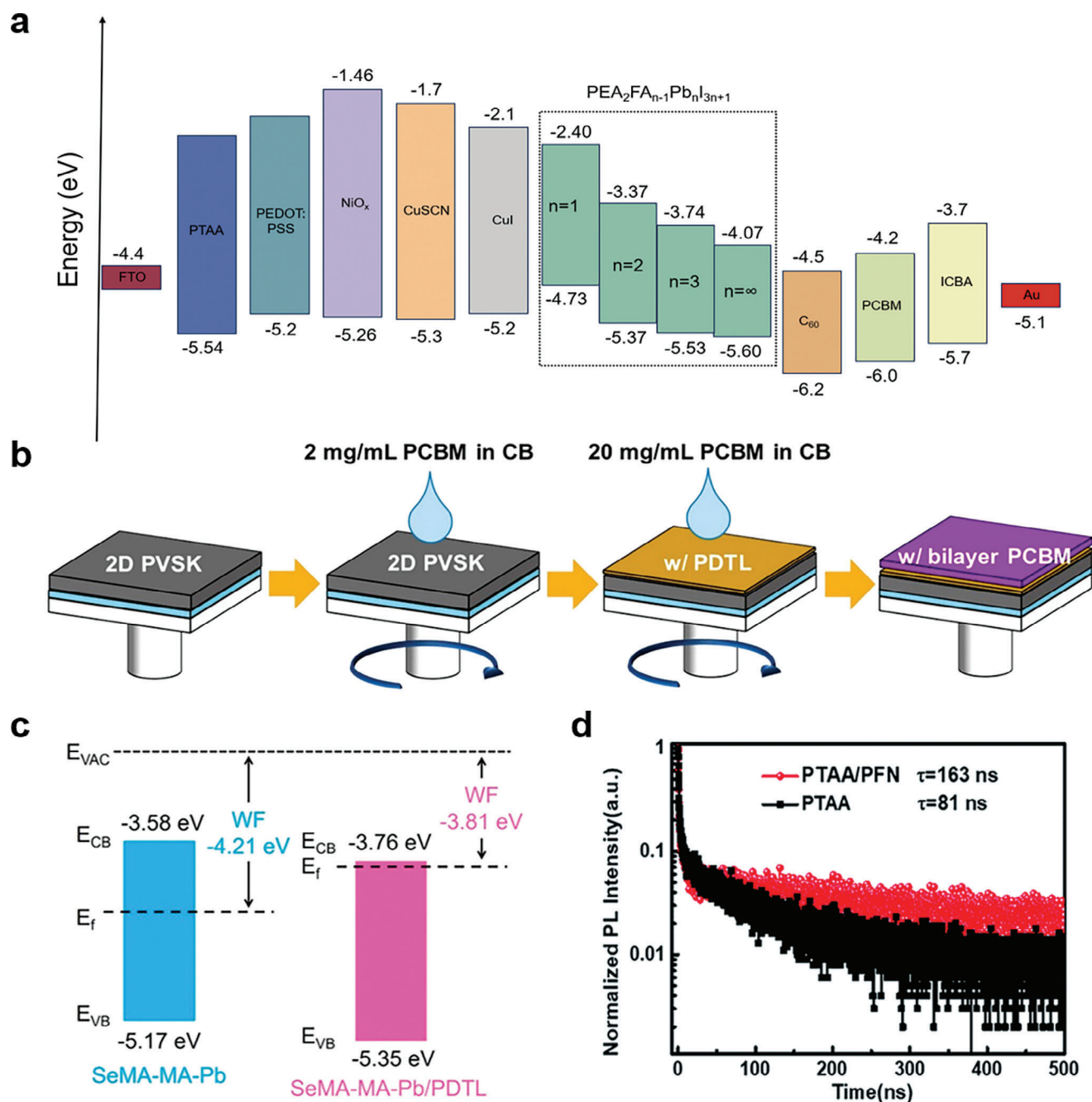


Figure 10. a) Energy levels of different materials acting as ETM and HTM in p-i-n structured perovskite PSCs. b) Procedures of PDLT of the PCBM method on 2DRP perovskite. Reproduced with permission.^[116] Copyright 2023, American Chemical Society. c) Schematic energy level diagrams of the 2D perovskite films before and after PDLT treatment. Reproduced with permission.^[116] Copyright 2023, American Chemical Society. d) Normalized TRPL spectra of the 2DRP PSCs based on PTAA and PTAA/PFN. Reproduced with permission.^[123] Copyright 2019, The Royal Society of Chemistry.

5.2.2. Buried Interface Modification

Similarly, buried interface modification in 2D perovskites facilitates the reduction of defect states at the interface and the regulation of the growth of the upper perovskite crystals.

By studying the difference between the vertical phase distribution of quasi-2D perovskite deposited on PEDOT: PSS and deposited on glass, Liu et al.^[132] found that the acidic components in

PEDOT: PSS would enter the precursor of perovskite and make it change from the original colloid to solution, and thus making it difficult to realize the ordered phase arrangement from small-*n* phase to large-*n* phase. As shown in Figure 11e, they introduced a 4-bromobenzene diazonium (BrB) tetrafluoroborate to modify the surface of PEDOT: PSS and finally obtained an ordered phase distribution conducive to charge dissociation and transport (Figure 11f). Liang et al.^[133] grew vertically oriented

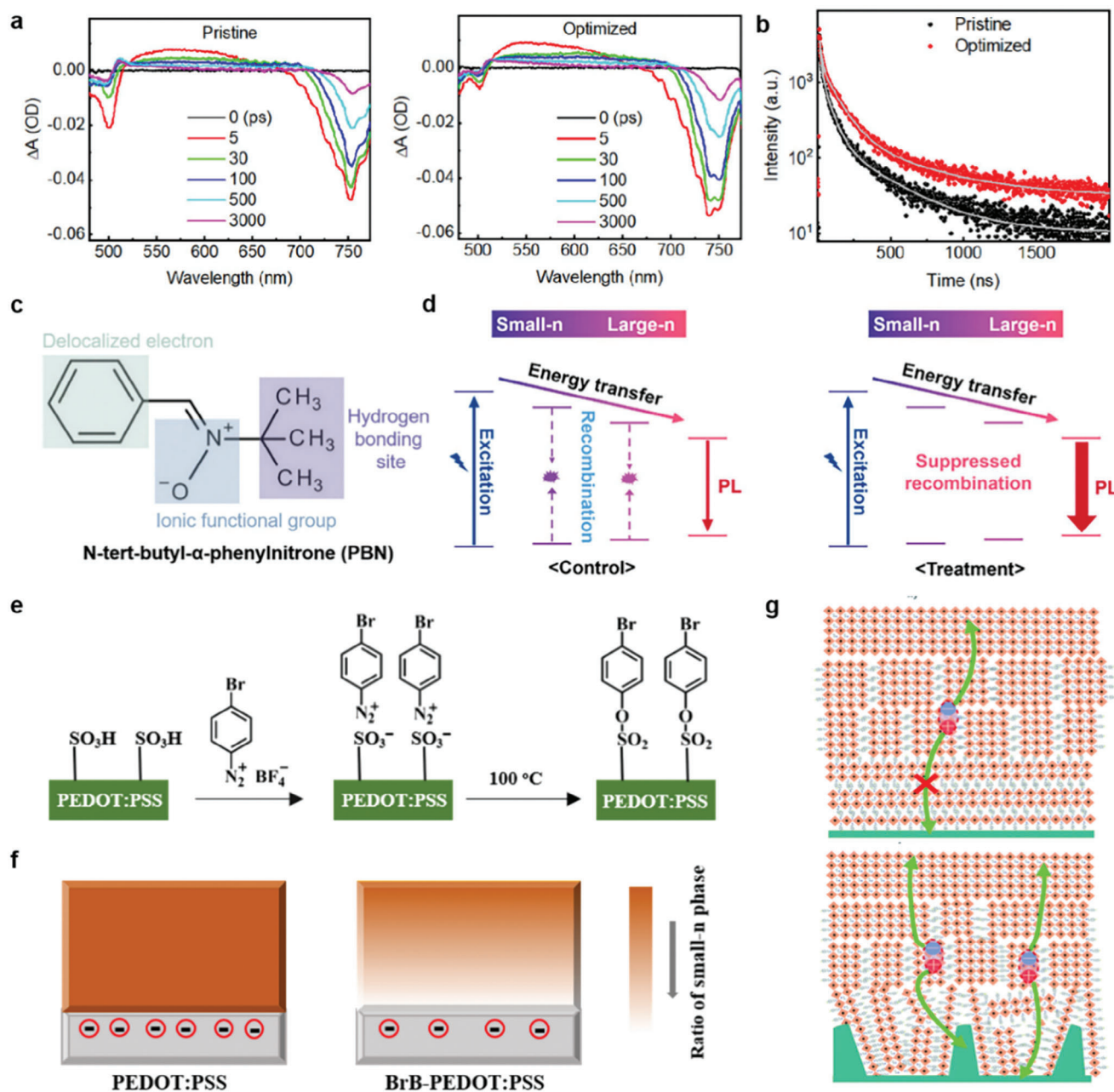


Figure 11. a) TA spectra of the pristine and optimized perovskite films, respectively (region size: $30 \times 30 \mu\text{m}$). Reproduced with permission.^[131] Copyright 2021, American Chemical Society. b) TRPL spectra of the pristine and optimized perovskite films, respectively (region size: $30 \times 30 \mu\text{m}$). Reproduced with permission.^[131] Copyright 2021, American Chemical Society. c) PBN molecular structure and possible function of each functional group, including the benzene group (green), nitron group (blue), and *t*-butyl group (purple). Reproduced with permission.^[108] Copyright 2023, Wiley-VCH. d) Schematic illustration of the energy-transport pathway in the control RPP material and the PBN-treated RPP material. Reproduced with permission.^[108] Copyright 2023, Wiley-VCH. e) The schematic of covalent Modification on PEDOT: PSS via a molecule of 4-bromobenzene diazonium tetrafluoroborate. Reproduced with permission.^[132] Copyright 2019, Springer Nature. f) Schematic of phase distribution (small-*n* to large-*n* phase from substrate side to the top side) in Q-2D perovskite films on PEDOT: PSS and BrB-PEDOT: PSS substrates. Reproduced with permission.^[132] Copyright 2019, Springer Nature. g) Schematic diagrams of the phase orientation, phase distribution, and carrier transport pathways in FPEA-planar and FPEA-lamella. Reproduced with permission.^[133] Copyright 2021, The Royal Society of Chemistry.

lamellar-NiO_x on a compact NiO_x HTL to overcome the limitation that the small-*n* phases at the bottom of the perovskite tend to be aligned in parallel by means of this mesoscopic structure (Figure 11g). The results showed that the interfacial resistance is significantly reduced and the photogenerated holes can be effectively collected by the bottom electrode due to the vertical alignment of the bottom phases.

There are very few reports with effective buried interface modifications in 2DRP perovskite cells. If more efficient and lower-cost buried interface modification materials can be developed, the *V*_{OC} and FF of 2DRP PSCs can undoubtedly be further improved.

6. Conclusions and Outlook

Compared with 3D perovskite, 2DRP perovskite has good structural, phase, and environmental stability. However, its PCE still lags behind 3D PSCs. A fundamental problem with RP perovskites is the delicate balance between hydrophobicity and charge conductivity. The introduction of hydrophobic organic spacer cations improves the stability of perovskites, but it sacrifices light absorption capacity and carrier transport efficiency, thus limiting device performance. In this review, we have analyzed the main reasons for the low current of 2DRP perovskite solar cells. We summarized thoroughly recent strategies to improve the charge transport in 2DRP perovskite thin films and devices.

The direct reason for the low current in 2DRP phase perovskite is the large binding energy of excitons, the difficulty of exciton dissociation, and the lack of free carriers. The quantum well distribution of different thicknesses, random orientation of crystals, and quantum and dielectric limits in 2DRP perovskites are key factors affecting the exciton binding energy. A reasonable perovskite bandgap plays a crucial role in the performance of solar cells. Approaches including *n* values regulation, A cation regulation and halogen anion regulation, have been discussed to adjust the perovskite bandgap. Moreover, perovskite structure regulation is analyzed to improve carrier transport in 2DRP perovskite films. The regulation of phase orientation, phase distribution, and the chemical interaction between amine molecules and inorganic layers in 2D perovskites can promote charge transport in 2DRP perovskite films. Apart from the carrier transport in perovskite film, ensuring both electrons and holes enter the external circuit along their respective transmission paths is crucial to the overall performance of perovskite devices. A specific energy level gradient and internal electric field between the various materials inside the device are necessary. Two aspects are discussed to improve the overall carrier transport level of the device. One is the appropriate electrode and carrier transport layer to ensure the level matching in n-i-p structured devices and p-i-n structured devices, and the other is the modification at different interfaces, including the perovskite surface and the buried interface.

However, to achieve single junction efficient and long-term operationally stable 2DRP PSCs, the study on 2DRP perovskite still needs more comprehensive and in-depth investigations. Two key points need to be further focused.

Firstly, selecting and designing suitable new spacer cations is crucial for achieving high device performance. By introducing chiral organic cation spacers or dipole organic cation spacers,

the structural symmetry of 2DRP is controlled, thereby introducing the Rashba effect, nonlinear optical properties, and endowing perovskite with a series of unique optoelectronic properties. Profoundly studying the mechanism of the influence of organic spacer ions on the performance of 2DRP perovskite is of great significance for higher PCE and stability. Nevertheless, a significant amount of workforce and material resources may be well-spent due to the diversity of spacer cation structures. Fortunately, machine learning may be an effective method for selecting promising organic cations for efficient and stable 2D perovskites, avoiding the process of repeated experimentation.

Secondly, the high-quality 2DRP perovskite film guarantees efficient carrier transport. Further study can be focused on the growth mechanism of 2DRP perovskite crystals, the development of effective film preparation technologies such as hot casting and steam treatment, and the combination of additive engineering and solvent engineering to regulate the crystallization kinetics of 2DRP perovskite, to obtain high-quality films with out-of-plane orientation and good phase distribution to improve the charge transfer process in the device.

In addition, for series stacked solar cells, pure phase 2DRP perovskite with a broader gap has excellent potential. At the same time, in order to realize the industrialization of 2DRP PSCs, the biggest problem is its large-scale preparation. In order to maintain the high PCE of large-scale 2DRP PSCs, it is essential to maintain the uniformity of 2DRP perovskite films. The performance reported to date needs to catch up to what is required for a large area 2DRP PSCs. The current thin film deposition method is a simple solution spin coating method, which is not conducive to preparing uniform large-area thin films. Therefore, to realize the commercial application of 2DRP perovskite devices, it is necessary to develop a new and efficient deposition method.

In summary, much work remains to be done for highly efficient and long-term operationally stable 2DRP perovskite-dominated single-junction photovoltaics. We hope this review can provide insight into relevant research. The in-depth study of 2DRP perovskite will eventually promote the commercialization of PSCs.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

bandgap regulation, charge carrier transporting, energy level matching, perovskite solar cells, two-dimensional Ruddlesden–Popper

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